



**SECP2523: DATABASE (WBL)**

**SEM I 2024/2025**

**DATABASE DESIGN DESCRIPTIONS: CONCEPTUAL  
DATABASE DESIGN**

**FOR <Online KADA System>**

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|-----------------------------|------------------|
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# **Introduction**

## **1.1 Overview about the company**

The current version of system documentation introduces the basis of KADA system. The introduction and specific requirements are included. The objective, scope, definitions, acronyms and abbreviations, references, and overview are all included in the introduction. To give a thorough grasp of the project, this portion of the system requirement specification also explains the persona, system characteristics such as the use case diagram, class diagram, state diagram, user stories, non-functional requirements and design constraints.

## **1.2 Problems**

### **1. Cumbersome Manual Registration Process**

All workflows are entirely manual, resulting in significant delays and inefficiencies. Users are often required to visit KADA branches repeatedly to complete processes, consuming substantial time and effort. Furthermore, the reliance on manual forms increases the likelihood of errors, causing additional delays in processing applications and other requests. Language barriers also pose a significant issue. All forms and documentation are available exclusively in Malay, limiting access for users who are not proficient in the language. This restriction significantly hinders KADA's ability to attract a diverse member base, as potential members who do not speak Malay may find the system inaccessible and challenging to navigate.

### **2. Lack of Clear Instructions**

The website provides limited instructions, making it difficult for users to understand essential processes such as registering as a member, applying for a loan, checking the status of their registration or application, and participating in agricultural programs. Without clear, step-by-step guidance, users are often left confused, leading to incomplete applications and missed opportunities to benefit from KADA's offerings.

### **3. Membership Cap**

With a current cap of only 500 members, many potential users are unable to access KADA's services, such as loans and agricultural programs. This limitation not only restricts the cooperative's growth but also reduces its ability to support a larger segment of the agricultural community.

### **4. Lack of Financial Transparency and Reporting**

Members are not provided with detailed financial reports, such as month-end or year-end profit and loss statements. This absence of transparency creates uncertainty and mistrust among members, as they have no means to assess the cooperative's financial health or understand how their contributions are being utilized.



Figure 1: Main Page of KADA

Borang Ini Diberi Percuma - *Diisi Oleh Pengusaha*

**BORANG PENDAFTARAN PESAWAH**

SKIM BAJA PADI KERAJAAN PERSEKUTUAN ( SBPKP )  
 INSENTIF PENGELUARAN PADI - PEMBAJAKAN ( IPKP )  
 INSENTIF PENGELUARAN PADI - INPUT PERTANIAN ( IIP )  
 INSENTIF PENINGKATAN PENGELUARAN BERAS NEGARA - ( IPB )

KEMENTERIAN PERTANIAN DAN INDUSTRI ASAS TANI MALAYSIA

Pertubuhan Peladang Kawasan No. Pendaftaran :  
 Tarikh Permohonan :  
 No. Kad Pengenalan :

**A. MAKLUMAT PENDAFTARAN**

**MAKLUMAT PENGUSAHA**

Nama Pengusaha / Syarikat :  
 No. Kad Pengenalan / No. Pendaftaran :  
 Alamat Surat Menyurat :  
 Poskod : Bandar : No. Tel / HP :

**KEAHLIAN PPK \***

Ahli PPK / Bukan Ahli PPK : Unit Peladang : No. Ahli :

**PENERIMAAN BANTUAN**

| Bil | Perkara                                     | Telah ( * ) Mendapat Bantuan |       | Jika " Ya " Nyatakan No. Pendaftaran | Keluasan Didaftar ( Hektar ) |
|-----|---------------------------------------------|------------------------------|-------|--------------------------------------|------------------------------|
|     |                                             | YA                           | TIDAK |                                      |                              |
| 1   | Subsidi Harga Padi                          |                              |       |                                      |                              |
| 2   | Skim Baja Padi Kerajaan Persekutuan         |                              |       |                                      |                              |
| 3   | Insentif Pengeluaran Padi - Upah Bajak      |                              |       |                                      |                              |
| 4   | Insentif Pengeluaran Padi - Input Pertanian |                              |       |                                      |                              |
| 5   | Insentif Peningkatan Pengeluaran Beras -    |                              |       |                                      |                              |

Figure 2: Manual Form Registration

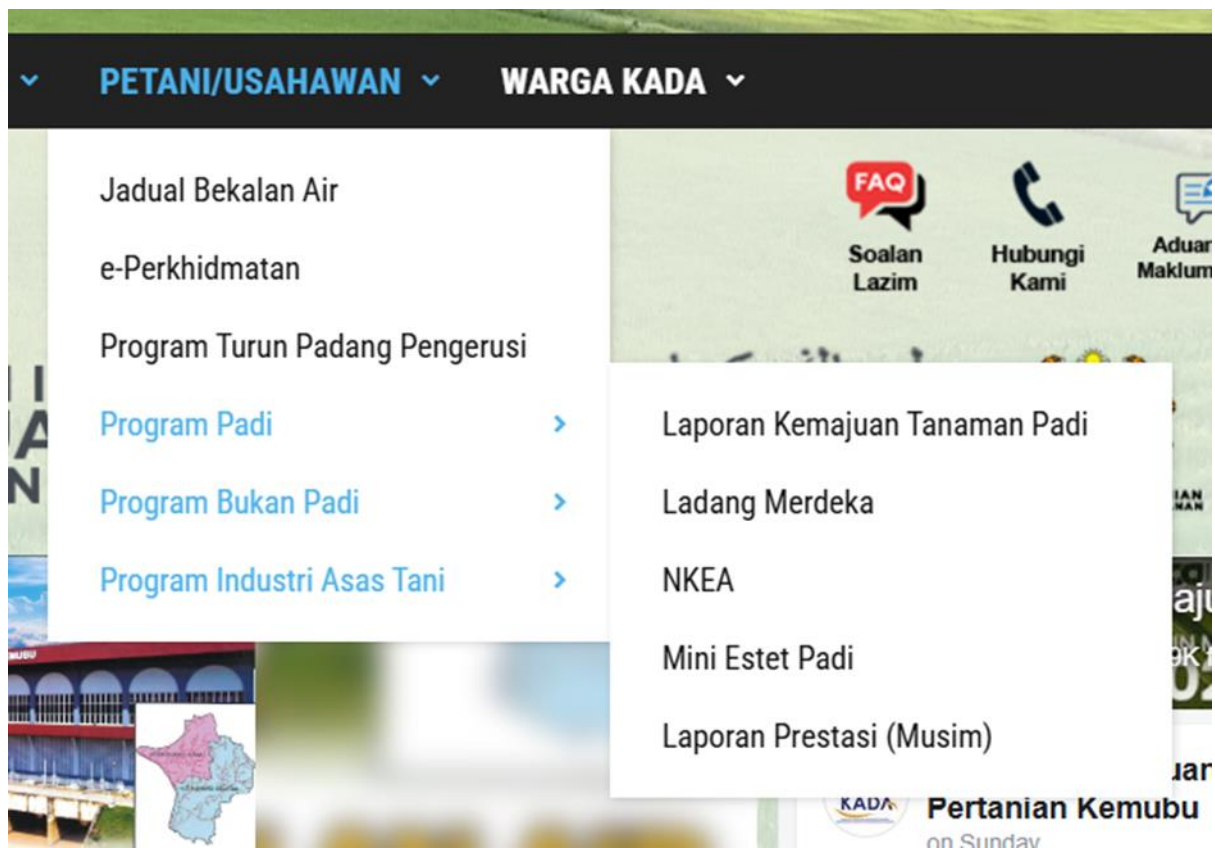


Figure 3: Google Analytic Missing



**Figure 4: Dazzling Selection**

### 1.3 Proposal

The system will be developed called online KADA System. It is a tracking system to track the development of the students in order to generate a complete resume that helps students in their future.

This system consists of 6 modules. These modules satisfied the fundamental requirements of the system. The modules included **Loan Application Module (LAM)**, **Program Application Module (PAM)**, **Saving Account Management Module (SAMM)**, **Loan Account Management Module (LAMM)**, **Financial Report Request Module (FRRM)**, and **Application Approval Module (AAM)**.

The **Loan Application Module** is one of the important components of the system that allows member to apply for loan. This module involves the interaction between Member and Admin. It includes key processes like submitting loan application, specifying loan details and monitoring application status. This module ensures a streamlined handling of loan requests then allowing members to track approval progress. Admin can view the application detail, oversee the approval process to ensure accurate evaluation of applications before finalizing decisions. This module facilitates an efficient loan application process while maintaining data accuracy and security.

The **Program Application Module** enables members to apply for programs offered by KADA. Actors here include Member and Admin. This module includes the submission of applications which consist of program names, descriptions, fees and approval timeline. Members can track the status of their applications while admin can manage and approve the applications. This module enhances transparency by maintaining the program status, where the member will get inform for the status. This module streamlines the program application process.

The **Saving Account Management Module** is focus on managing member's saving account. The actor for this module is Member. This module handles processes like account creation, deposit, and withdrawal. It ensures member can access to their account details, like the share capital, fixed savings, interest rates and transaction history. This module provides effective management of savings, where member can make informed decisions about their account.



The **Loan Account Management Module** is responsible for managing loan accounts for member. The actor for this module is Member and Admin. Members can view their loan details, including total loan available, loan installment and payment history. This module provides real-time access to loan account information, where supports member maintain a structured loan management process.

The **Financial Report Request Module** focus on generating and managing financial reports for members. Member can request their financial reports like monthly or yearly summaries on their finance status where generated based on their account and transaction data. This module will provides detailed financial insights that enable member to understand their financial standing and make informed decisions.

The **Application Approval Module** is also an important part of the system that ensures proper evaluation and approval of Member applications. The actors in this module is Admin. The admin can access the application and update the approval status. By maintaining structured and secure approval workflows, this module promotes consistency and fairness in application processing.

## 1.4 Definitions, Acronyms and Abbreviations

| No. | Acronym/Abbreviations | Definition                        |
|-----|-----------------------|-----------------------------------|
| 1   | KADA                  | Lembaga Kemajuan Pertanian Kemubu |
| 2   | Admin                 | Administer                        |
| 3   | PAM                   | Program Application Module        |
| 4   | LAM                   | Loan Application Module           |
| 5   | SAMM                  | Saving Account Management Module  |
| 6   | FRRM                  | Financial Report Request Module   |
| 7   | AAM                   | Application Approval Module       |

# Database System Requirement

## 2.1 System Overview

The proposed Online KADA System is designed to streamline and digitize the operations of the Kemubu Agricultural Development Authority (KADA). The system eliminates inefficiencies in manual processes by integrating six basic modules: Loan Application, Program Application, Saving Account Management, Loan Account Management, Financial Report Request, and Application Approval. Each module facilitates key processes like membership registration, loan applications, program participation, account management, and financial reporting. The Online KADA System uses a centralized database system to provide safe, accurate, and real-time data access to both members and admin, allowing for more efficient decision-making, transparency and a better user experience.

## 2.2 Module <Loan Application>

JOANNE CHING YIN XUAN is the person in charge of documenting this module.

### 2.2.1 Module Description

This module will involve Members and Admins. There are 3 use cases inside this module which are UC01: Apply Membership, UC02: View Application Status, UC003: Login, UC004: Apply Loan Application, and UC11: Approve Loan Application

UC01: Apply Membership

- A user accesses the membership application page.
- The user fills out the online form with personal and required details.
- The system validates the inputs and stores the application for review.

UC02: View Application Status

- Members can track the status of their loan application whether it is Pending, Approved, or Rejected

UC03: Login

- The system provides a login interface that is accessible to members and the admin.
- Member shall enter their username and password for authentication.
- System will validate the user credentials against the stored database.
- Upon successful validation, the system shall grant access to the respective page

#### UC04: Apply Loan Application

- Members can submit loan applications through the system
- The application form includes fields such as Loan Amount, Purpose of Loan, and the desired repayment period
- The system performs validation to ensure all required fields are completed and loan amount and repayment comply with eligibility criteria
- Once submitted, the application is saved in the system for further review and approval by the admin

#### UC11: Approve Loan Application

- The admin would be able to update the status of the program application.

### **2.2.2 Transaction Requirement: Data Entry**

#### UC01: Apply Membership:

- Member input personal details such as name, contact information, and address.
- Provide a unique username and password for account creation.
- Upload supporting documents.

#### UC03: Login

- Members and Admins enter a valid username and password for authentication.

#### UC04: Apply Loan Application

- Input the following fields in the loan application form:
  - o Loan amount
  - o Purpose of the loan
  - o Desired repayment period
  - o Provide any additional required details, such as income information

#### UC11: Approve Loan Application

- Admin updates the application status to Approved, Pending, or Rejected.

## 2.2.3 Transaction Requirement: Data Update/Deletion

### UC01: Apply Membership

- Data Update:
  - o Members can update personal details such as name, contact information, and address after account creation.
  - o Update or reset the username or password if forgotten or compromised.
- Data Deletion:
  - o Admins can delete invalid or duplicate membership applications.

### UC02: View Application Status

- Data Update:
  - o Members can update their loan application details if the status remains Pending.
- Data Deletion:
  - o Loan applications may be withdrawn by members or deleted by admins if found incomplete or invalid.

### UC03: Login

- Data Update:
  - o Members and admins can reset or change their password if required.
- Data Deletion:
  - o Remove login credentials for inactive members or resigned admins.

### UC04: Apply Loan Application

- Data Update:
  - o Members can edit their loan application (e.g., loan amount or repayment period) before the application is processed.  
Admins can correct errors in member-submitted loan applications if requested.
- Data Deletion:
  - o Admins can delete loan applications that do not meet eligibility criteria or are flagged as fraudulent.

### UC11: Approve Loan Application

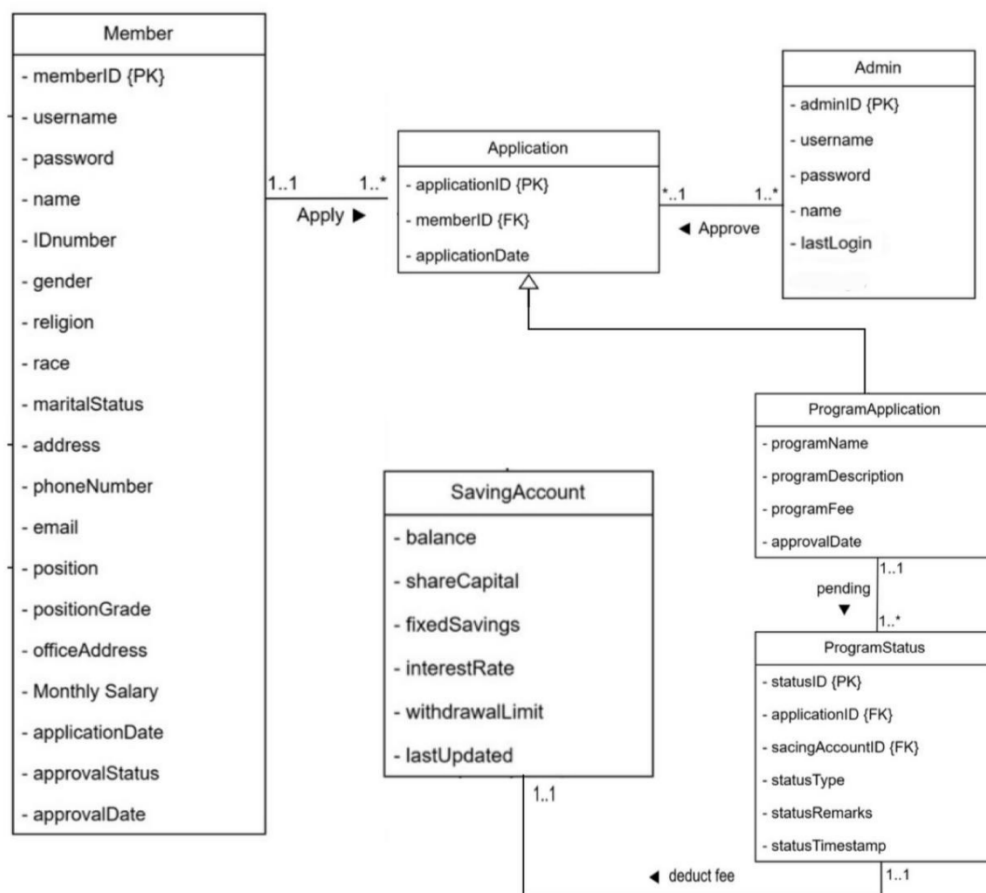
- Data Update:
  - o Admins can update the status of loan applications (e.g., change status from Pending to Approved or Rejected).
  - o Add or modify comments/remarks related to the application status.

- Data Deletion:
  - o Remove loan applications that have been rejected or withdrawn by the member.

## 2.2.4 Transaction Requirement: Data Queries

1. View the list of loan applications with details such as loan type, loan amount, loan period, and status.
2. View a history of loan applications for a specific member, including approved, pending, or rejected loans

## 2.2.5 Local Conceptual Data Model (ERD) for Module 1



## **2.3 Module <Saving Account Management>**

Lim Yu Han is the person in charge of documenting this module.

### **2.3.1 Module Description**

This module will involve actors Admin and Member. There is 5 use case inside this module: UC01: Apply membership, UC02: View application status, UC03: Login, UC06: Manage Saving Account and UC10: Approve Membership Application.

#### **UC01: Apply membership**

1. A user accesses the membership application page.
2. The user fills out the online form with personal and required details.
3. The system validates the inputs and stores the application for review.

#### **UC02: View application status**

1. Users can view the status of their membership, loan or program application.
2. The system displays the status as approved, rejected or pending.

#### **UC03: Log in**

1. Members and admins log in with their unique username and password.
2. Upon successful login:
  - Members are redirected to member dashboard.
  - Admins are redirected to the admin dashboard.
3. The system ensures secure authentication for all users.

#### **UC10: Approve membership application**

1. The admins log in the system.
2. Admin views a list of pending membership applications.
3. Admin selects an application to review and updates its status.
4. The system notifies the user of the decision and stores the updated status.

## UC06: Manage Saving Account

1. Each member is assigned a saving account upon registration.
2. Members can deposit and withdraw funds within limits.
3. Members can also view account details (balance and transaction history)
4. Admins can monitor saving account activities for all members.
5. Admins can also adjust account settings like withdrawal limits.
6. Every transaction is logged with a timestamp, amount and optional description.
7. Admins can generate reports for analysis and decision-making.

### **2.3.2 Transaction Requirement: Data Entry**

1. Members must enter transaction amounts (deposit and withdrawal).
2. Transaction descriptions are optional.
3. Each transaction is automatically allocated a unique ID and timestamp.
4. Input validation guarantees data accuracy and integrity.

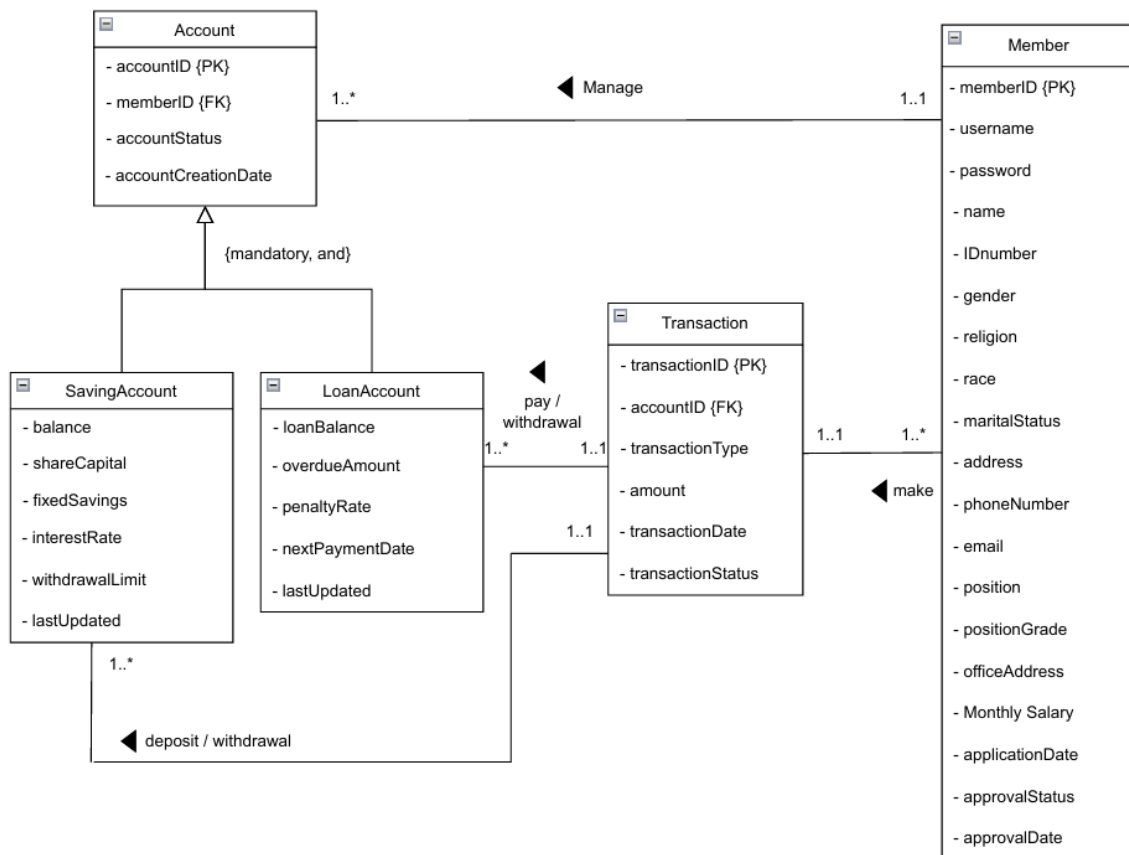
### **2.3.3 Transaction Requirement: Data Update/Deletion**

1. Only administrators can change account information, including withdrawal limits.
  2. Only administrators can reverse/delete transactions, such as undoing or removing incorrect entries, with all actions recorded.

### **2.3.4 Transaction Requirement: Data Queries**

1. Members can view account details (balance and limits)
2. Transaction history reports have filters for date, type and amount.
3. Admins can generate detailed summaries for deposits and withdrawals.
4. A search feature enables queries by member ID and transaction ID.

## 2.3.5 Local Conceptual Data Model (ERD) for Module 2



## 2.4 Module <Program Application>

JOANNE CHING YIN XUAN is the person in charge of documenting this module.

### 2.4.1 Module Description

This module will involve Members and Admins. There are 3 use cases inside this module which are UC01: Apply Membership, UC02: View Application Status, UC003: Login, UC004: Apply Loan Application, and UC11: Approve Loan Application

UC03: Login

- The system provides a login interface that is accessible to members and the admin.
- Member shall enter their username and password for authentication.
- System will validate the user credentials against the stored database.
- Upon successful validation, the system shall grant access to the respective page



#### UC01: Apply Membership

- A user accesses the membership application page.
- The user fills out the online form with personal and required details.
- The system validates the inputs and stores the application for review.

#### UC05: Apply Program

- Members can submit loan applications through the system
- The application form includes fields such as Loan Amount, Purpose of Loan, and the desired repayment period
- The system performs validation to ensure all required fields are completed and loan amount and repayment comply with eligibility criteria
- Once submitted, the application is saved in the system for further review and approval by the admin

#### UC02: View Application Status

- Members can track the status of their program application whether it is Pending, Approved, or Rejected
- If the program application is approved, the program fee will be deducted from the Member's Saving Account.

#### UC11: Approve Program Application

- The admin would be able to update the status of the program application.

### **2.4.2 Transaction Requirement: Data Entry**

#### UC03: Login

- Members and Admins enter a valid username and password for authentication.

#### UC05: Apply Program

- Input the following fields in the loan application form:
  - o Loan amount
  - o Purpose of the loan
  - o Desired repayment period
  - o Provide any additional required details, such as income information

#### UC12: Approve Loan Application

- Admin updates the application status to Approved, Pending, or Rejected.

### **2.4.3 Transaction Requirement: Data Update/Deletion**

#### **UC02: View Application Status**

- Data Update:
  - o Members can update their loan application details if the status remains Pending.
- Data Deletion:
  - o Loan applications may be withdrawn by members or deleted by admins if found incomplete or invalid.

#### **UC03: Login**

- Data Update:
  - o Members and admins can reset or change their password if required.
- Data Deletion:
  - o Remove login credentials for inactive members or resigned admins.

#### **UC04: Apply Loan Application**

- Data Update:
  - o Members can edit their loan application (e.g., loan amount or repayment period) before the application is processed.
  - o Admins can correct errors in member-submitted loan applications if requested.
- Data Deletion:
  - o Admins can delete loan applications that do not meet eligibility criteria or are flagged as fraudulent.

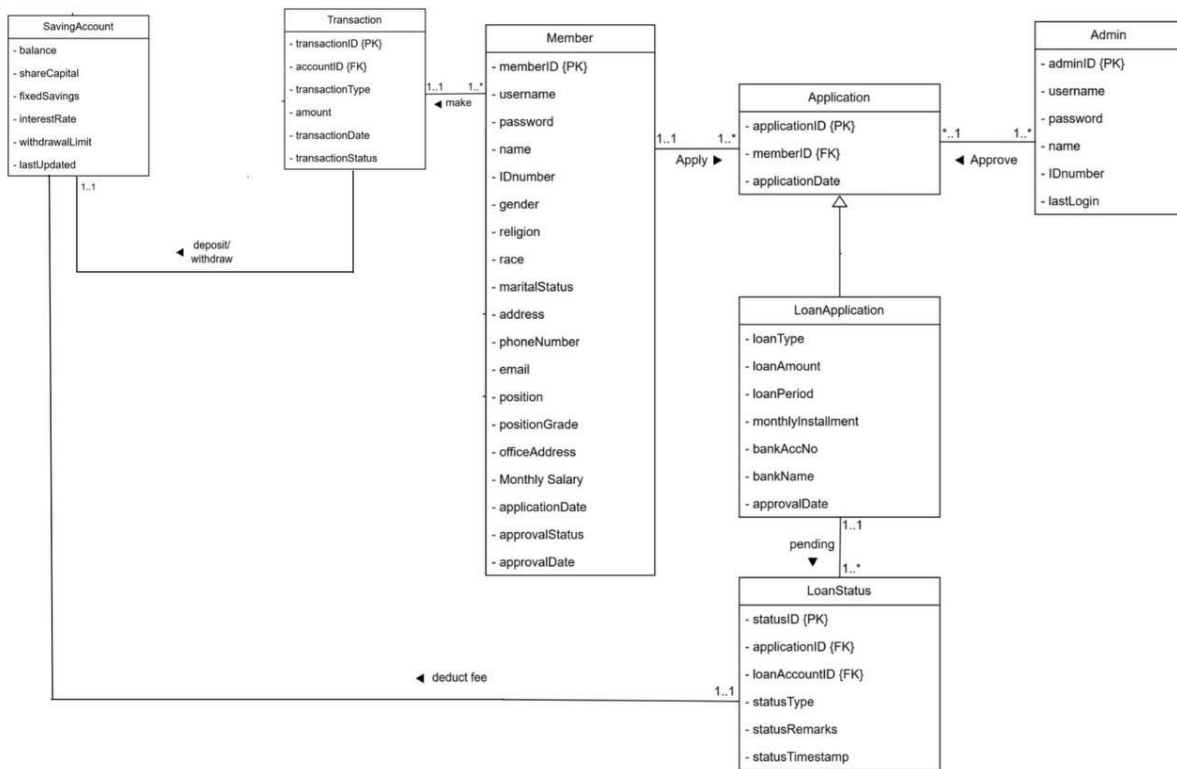
#### **UC12: Approve Loan Application**

- Data Update:
  - o Admins can update the status of program applications (e.g., change status from Pending to Approved or Rejected).
  - o Add or modify comments/remarks related to the application status.
- Data Deletion:
  - o Remove program applications that have been rejected or withdrawn by the member.

### **2.4.4 Transaction Requirement: Data Queries**

1. View the list of program applications with details such as program name and purpose, and status.
2. View a history of program applications for a specific member, including approved, pending, or rejected program

## 2.4.5 Local Conceptual Data Model (ERD) for Module 3



## **2.5 Module <Loan Account Management>**

Choh Jing Yi is the person in charge of documenting this module.

### **2.5.1 Module Description**

This module will involve Members and Admin. There are five use cases inside this module which are UC02: View Application Status, UC04: Apply Loan Application, UC08 View Finance Report, UC09 Admin View Application List and UC11 Approve Loan Application.

#### **UC02: View application status**

1. Users can view the status of their loan application.
2. The system displays the status as approved, rejected or pending.

#### **UC04 Apply Loan Application:**

1. The member accesses the loan application page from the member dashboard.
2. The member fills in the loan application form, including details such as loan type, loan amount, loan period and monthly income.
3. The member uploads the required supporting documents for example proof of income.
4. The system validates the form and document uploads for completeness and accuracy.
5. The system stores the application and marks it as "Pending" for admin review.
6. The member can check their status whether the loan status is Approved or Reject.

#### **UC08 View Financial Status:**

1. The member navigates to the "Financial Status" page.
2. The system displays the following real-time financial details such as loan balance, loan period, overdue amount, next payment date and monthly installment details.
3. The member can filter or search for specific details, such as loan type or period.
4. The system ensures that the displayed data is up-to-date and accurate.

#### **UC09 Admin View Application List:**

1. The admin can sort or filter the list based on criteria such as loan type or applicant name.
2. The admin selects a specific application to review further.

#### **UC11 Approve Loan Application:**

1. The admin logs into the system and navigates to the "Loan Applications" section.
2. The admin views a list of pending loan applications.
3. The admin selects a specific application to review.
4. The system displays the applicant's loan details, supporting documents, and eligibility criteria.

5. The admin evaluates the application and provides a decision whether approve or reject.
6. The system updates the application's status
7. If approved, the loan account is created and linked to the member's account.

## **2.5.2 Transaction Requirement: Data Entry**

UC04 Apply Loan:

Members submit loan applications by filling out forms with details such as loan type, amount, loan period, monthly income, and uploading supporting documents.

UC09 Admin View Application List:

Admins review and manage loan applications by filter, sort, or search through the list of applications.

UC11: Approve Loan Application

Admins are responsible for entering loan approval decisions, such as marking applications as "Approved" or "Rejected."

## **2.5.3 Transaction Requirement: Data Update/Deletion**

UC02: View Application Status

- Data Update:
  - o Members can update their loan application details if errors are identified or additional information is required before the application is reviewed by admins.
- Data Deletion:
  - o Loan applications may be withdrawn by members or deleted by admins if found incomplete or invalid.

UC04: Apply Loan Application

- Data Update:
  - o Members can update their application before it is reviewed (e.g., uploading missing documents).
  - o Admins can update loan application details if required. (e.g., correcting errors in loan amount, loan period, or personal details provided by members)
- Data Deletion:
  - o Admins can delete invalid or duplicate loan applications if identified during the review process.

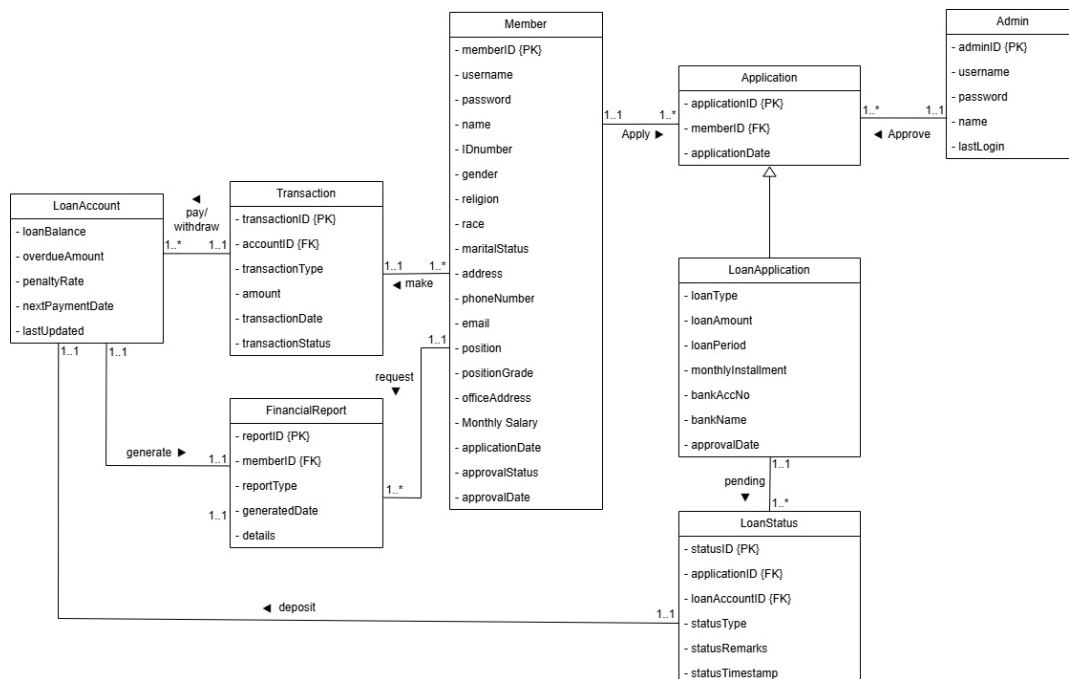
## UC11: Approve Loan Application

- **Data Update:**  
Admins can update the loan application status to "Approved" or "Rejected" based on the review.  
Update remarks or notes about the approval or rejection decision.
- **Data Deletion:**
  - Delete loan applications that are invalid, duplicate, or fraudulent after thorough verification.

### 2.5.4 Transaction Requirement: Data Queries

- Members can view the application status. The system retrieves the status of a member's loan application and displays whether it is approved, rejected, or pending.
- Members can view loan account details in financial report such as loan balance, loan period, overdue amount, next payment date, and monthly installment details.
- Admins can retrieve pending loan applications, including the application ID, member financial details (e.g., monthly income and credit history), loan application details (e.g., loan type, amount, and period), and supporting documents, enabling them to evaluate and approve or reject the applications.

### 2.5.5 Local Conceptual Data Model (ERD) for Module 5



## **2.6 Module <Financial Report Request>**

Elijah She Yu Sheng is the person in charge of documenting this module.

### **2.6.1 Module Description**

This module involves two actors: Admin and Member. It includes the following use cases:

#### **UC07: Create Financial Report**

Members can generate financial reports (monthly or annual) for their savings, loans, and transactions.

Members select the report type (monthly or annual) and specify the date range.

The system retrieves relevant data from the database, including transactions, loans, and contributions.

A downloadable report is generated in the chosen format (PDF/Excel).

#### **UC08: View Financial Status**

Members can view their current financial status, including outstanding loan balances, due dates, and savings contributions.

Members log in and access the "Financial Status" section.

The system displays real-time data on savings, loans, and other financial metrics.

#### **UC09: Admin View Application List**

Admins can query and filter applications to generate relevant financial summaries.

Admins access a summary of financial applications and related data.

The system enables filtering by application type, date, and status for detailed analysis.

#### **UC13: Manage Website (Financial Report Policies)**

Admins can manage and update financial reporting policies and configurations.

Admins update financial reporting policies via the system interface.

The system ensures changes are logged and visible in real-time to relevant user.

## **2.6.2 Transaction Requirement: Data Entry**

### **UC07: Create Financial Report**

Members enter the desired report type (monthly or annual).  
Specify the date range for the report.  
Select additional parameters, such as including savings, loans, or both.

### **UC08: View Financial Status**

Members log in to view real-time financial data.  
Data fields displayed include current balance, outstanding loans, and due dates.

### **UC09: Admin View Application List**

Admins input filters to query applications by type (membership, loan, program).  
Admins provide specific time ranges or applicant details for more refined searches.

### **UC13: Manage Website**

Admins input policy updates, including interest rates and reporting formats.  
Admins provide justifications for significant changes, which are logged in the system.

## **2.6.3 Transaction Requirement: Data Update/Deletion**

### **UC07: Create Financial Report**

#### **Data Update:**

Members can modify pending report requests (e.g., change date range or report type).  
Admins can update report generation settings to address errors or include additional data.

#### **Data Deletion:**

1. Members can delete their pending report requests.  
Admins can delete failed or erroneous reports, ensuring logs are maintained.

### **UC08: View Financial Status**

#### **Data Update:**

Members can refresh displayed data to ensure the most up-to-date financial status.  
Admins can update financial metrics for corrections, such as adjusted loan balances.

### **UC09: Admin View Application List**



#### Data Update:

Admins can update the status or remarks of membership, loan, or program applications.  
Admins can batch process updates for multiple applications.

#### Data Deletion:

Admins can remove duplicate or invalid applications from the system.

#### UC13: Manage Website

#### Data Update:

Admins can revise financial policies or configurations, such as default report formats.  
Add or edit metadata for reports, such as titles and version details.

#### Data Deletion:

Admins can remove outdated policies and replace them with updated versions.

### **2.6.4 Transaction Requirement: Data Queries**

#### UC07: Create Financial Report

Members can query generated reports by type, date, or status (e.g., pending, completed).  
Search for specific reports using report ID or date range.

#### UC08: View Financial Status

Members can query their current financial details, such as total savings or outstanding loans.  
Admins can retrieve detailed financial summaries for specific members.

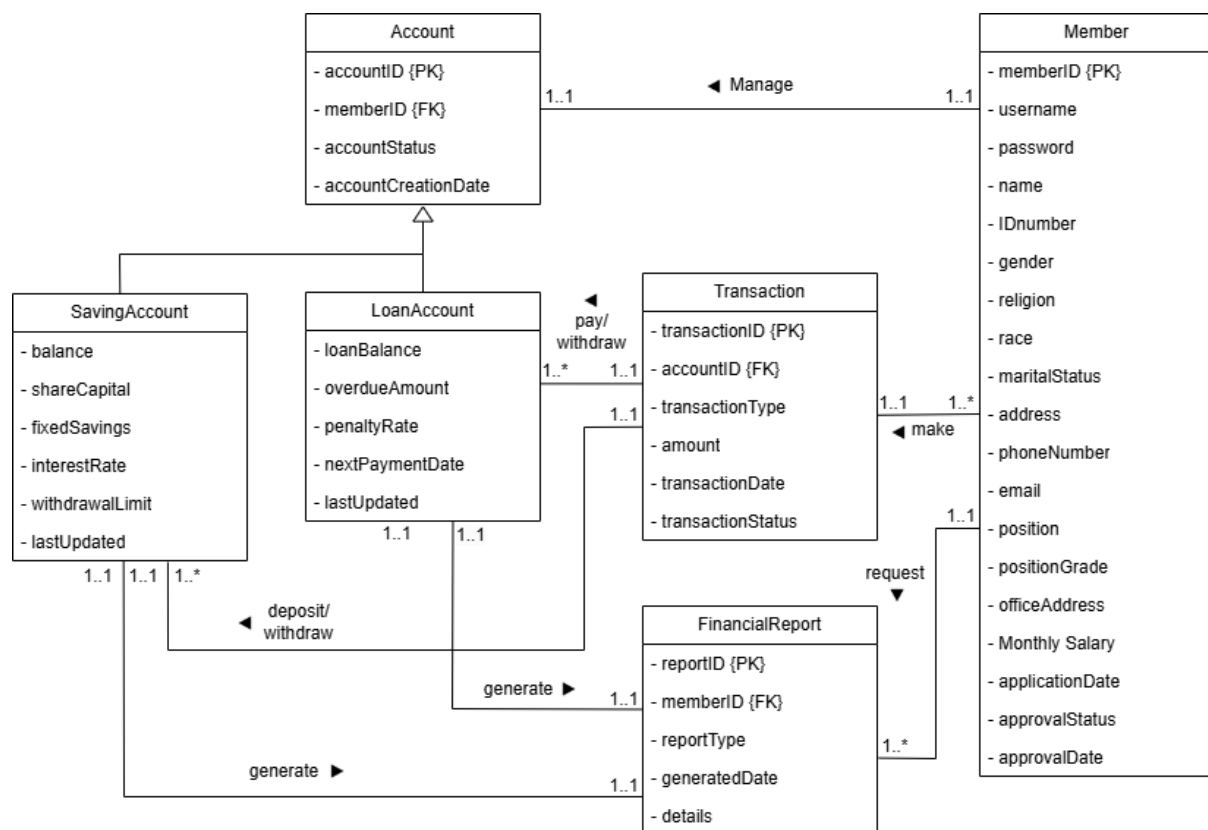
#### UC09: Admin View Application List

Admins can query a list of applications by type (membership, loan, program) and status (approved, pending, rejected).  
Admins can filter applications by date, applicant name, or ID for detailed analysis.

#### UC13: Manage Website

Admins can view and audit logs of policy changes related to financial reports.  
Query historical report settings for compliance and transparency.

## 2.6.5 Local Conceptual Data Model (ERD) for Module 5



## **2.7 Module <Application Approval Module>**

Lee Yin Shen is the person in charge of documenting this module.

### **2.7.1 Module Description**

The Application Approval Module facilitates the review and approval process for loan and program applications. It involves the two actors, Members and Admins. Three main use cases applied in this module: UC08: Review Application List, UC09: Approve Loan Application, UC10: Approve Program Application.

#### **1. UC08: Review Application List**

- Admins can view a list of loan and program applications submitted by members.
- Details such as application type, member details and status will be display.

#### **2. UC09: Approve Loan Application**

- Admins review loan applications to ensure it is compliance with eligibility criteria.
- Admins can approve, reject, then update the status for an application.
- Admins can provide remarks and timestamps.

#### **3. UC10: Approve Program Application**

- Admins review program applications to ensure it is compliance with eligibility criteria.
- Admins can approve, reject, then update the status for an application.
- Admins can provide comments and timestamps.

### **2.7.2 Transaction Requirement: Data Entry**

#### **1. UC08: Review Application List**

- Admins input the search criteria to filter applications by Member ID, status or application type.
- Admins remarks during the review process (if necessary).

#### **2. UC09: Approve Loan Application**

- Admins update loan approval status (Approved, Pending, Rejected).

- Admins update remarks like missing documentation.
- Admins update approval timestamp.

### **3. UC10: Approve Program Application**

- Admins record program applications status.
- Admins can approve, reject, then update the status for an application.
- Admins can provide comments and timestamps.

## **2.7.3 Transaction Requirement: Data Update/Deletion**

### **1. UC08: Review Application List**

- Data Update:
  - Admins can modify application details for consistency.
- Data Deletion:
  - Admins can delete invalid or duplicate applications.

### **2. UC09: Approve Loan Application**

- Data Update:
  - Admins can modify application details for consistency.
- Data Deletion:
  - Admins can delete invalid or duplicate applications.

### **3. UC10: Approve Program Application**

- Data Update:
  - Admins revise approval details if requested by the Member or upon identifying errors.
- Data Deletion:
  - Admin can delete program applications that fail criteria.

## **2.7.4 Transaction Requirement: Data Queries**

### **1. Retrieve Application List**

- Admins retrieve a filtered list of applications by:

- Applications type (Loan/Program)
- Status (Pending/Approved/Rejected)
- Member details

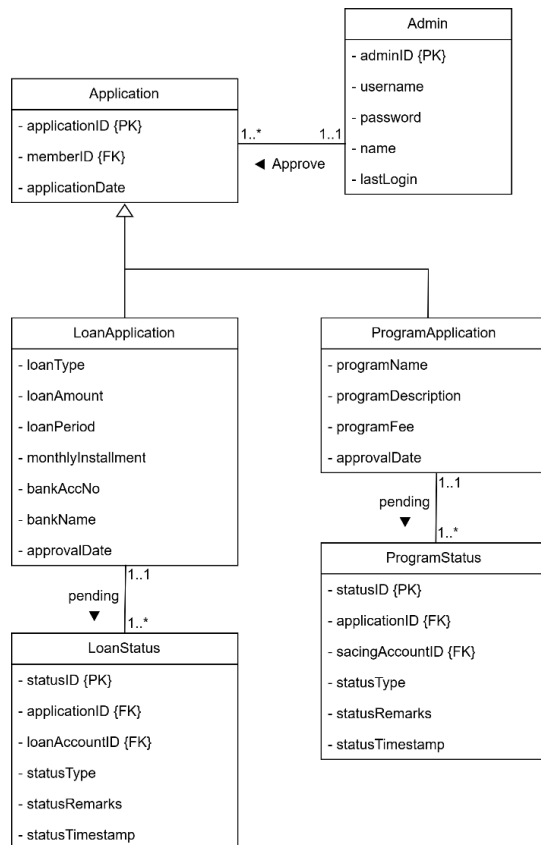
## 2. Track Application History

- Admins view a historical record of:
  - Applications by a specific member.
  - Status updates, including timestamps and remarks.

## 3. Generate Approval Reports

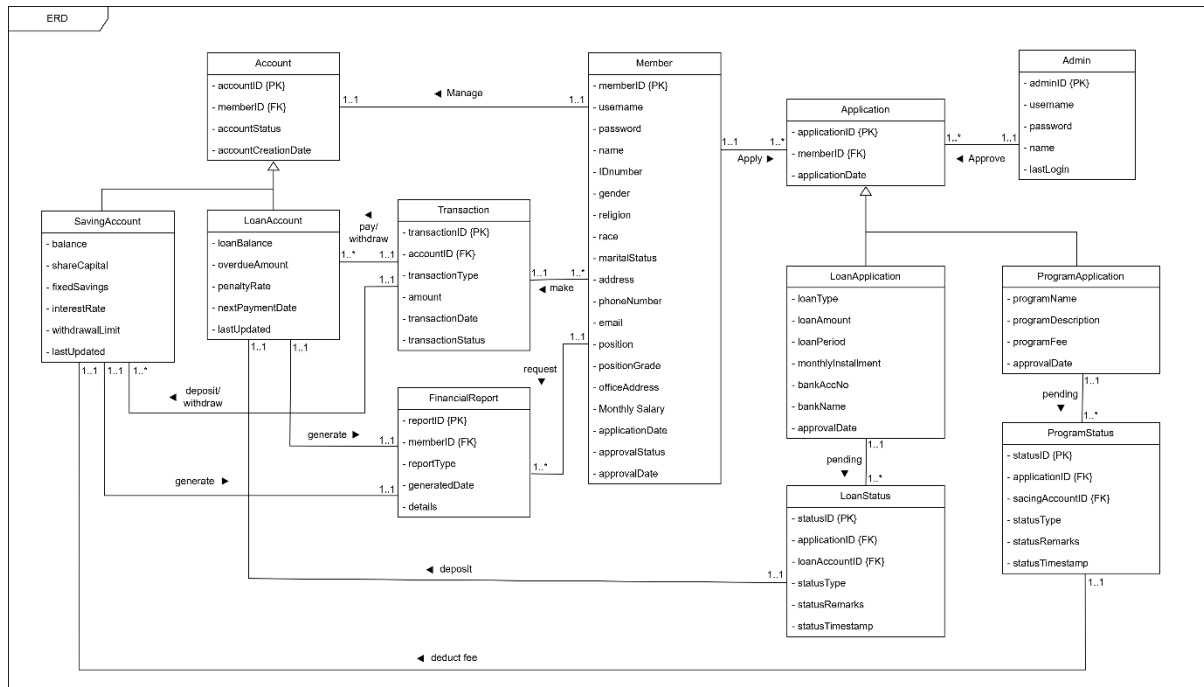
- Admins generate reports on:
  - Total applications received.
  - Approved and rejected Applications.

### 2.7.5 Local Conceptual Data Model (ERD) for Module 6

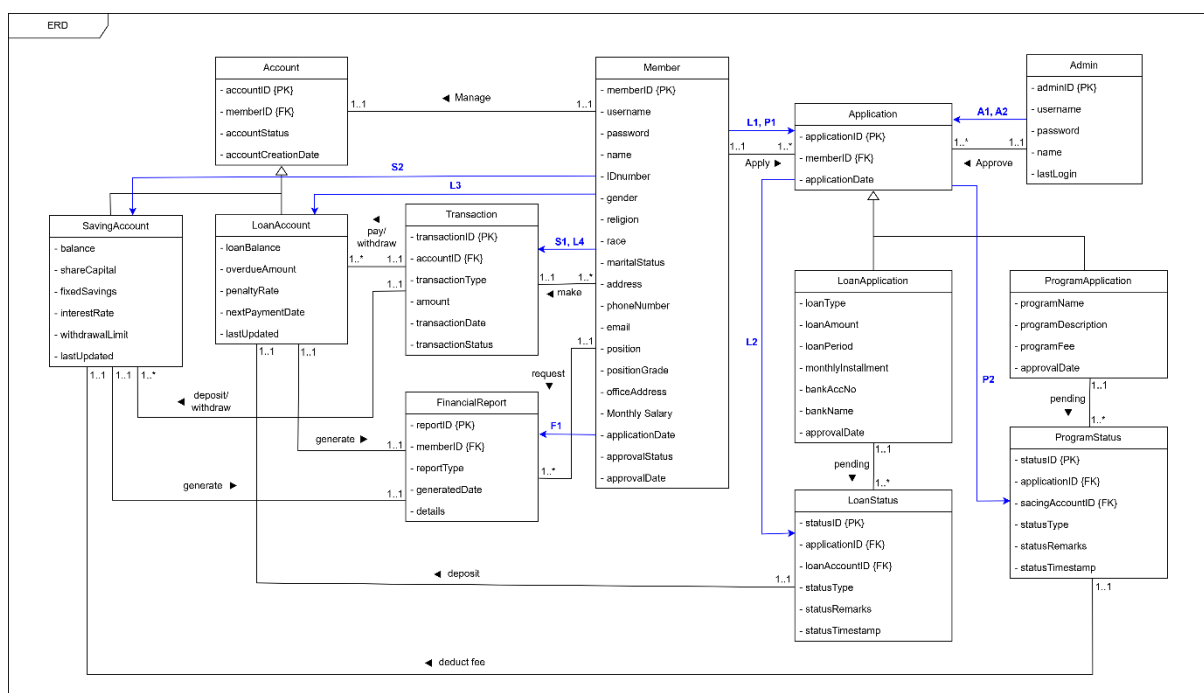


# Global Conceptual Data Model (Global ERD)

## 3.1 Global Conceptual Design (ERD)



## 3.2 Verified Global Conceptual Design (Verified ERD)



| Symbol | Transaction                                                                                                                                                                                                  |
|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| L1     | Submit Loan Application: Members submit loan applications with details such as loan type, amount, and period ( <i>LoanApplication</i> , <i>LoanStatus</i> ).                                                 |
| L2     | View Loan Application Status: Members view the status of their loan applications (Pending, Approved, Rejected) ( <i>LoanStatus</i> ).                                                                        |
| L3     | View Loan Account Details: Members view loan account details, including balances, overdue amounts, and payment schedules ( <i>LoanAccount</i> ).                                                             |
| L4     | Repay Loan: Members make loan repayments, which update their loan account balances and are recorded as transactions ( <i>LoanAccount</i> , <i>Transaction</i> ).                                             |
| P1     | Submit Program Application: Members apply for a program, providing details like program name, description, and fee ( <i>ProgramApplication</i> , <i>ProgramStatus</i> ).                                     |
| P2     | View Program Application Status: Members check the status of their program applications ( <i>ProgramStatus</i> ).                                                                                            |
| S1     | Deposit/Withdraw from Savings: Members perform deposits or withdrawals from their saving accounts ( <i>SavingAccount</i> , <i>Transaction</i> ).                                                             |
| S2     | View Saving Account Details: Members view their savings account details, including balance, fixed savings, and transaction history ( <i>SavingAccount</i> ).                                                 |
| F1     | Generate Financial Report: Members request financial reports for their accounts and transactions ( <i>FinancialReport</i> , <i>Transaction</i> , <i>Account</i> ).                                           |
| A1     | Review Applications: Admins filter and view lists of submitted loan and program applications by status, type, or member details ( <i>Application</i> , <i>LoanApplication</i> , <i>ProgramApplication</i> ). |
| A2     | Approve/Reject Applications: Admins evaluate and update the status of applications (loan or program) ( <i>LoanStatus</i> , <i>ProgramStatus</i> ).                                                           |

#### Module-to-Transaction Mapping:

1. **Loan Application Module:** L1, L2
2. **Program Application Module :** P1, P2
3. **Saving Account Management Module:** S1, S2
4. **Loan Account Management Module:** L3, L4
5. **Financial Report Request Module:** F1
6. **Application Approval Module:** A1, A2

### 3.3 Data Dictionary for the global conceptual ERD.

#### 3.3.1 Identify Entity

| Entity Name            | Description                                                                                                     | Aliases                      | Occurrence                                                   |
|------------------------|-----------------------------------------------------------------------------------------------------------------|------------------------------|--------------------------------------------------------------|
| User                   | Represents a user in the system, which could be an admin or a member.                                           | Account Holder, User         | Occurs when a user is created in the system.                 |
| Member                 | Represents an individual who has registered and been approved as a cooperative member.                          | Applicant, Member            | Occurs when a user applies and is approved for membership.   |
| Admin                  | Represents the administrator who manages applications, system info, and website content.                        | Admin, Manager               | Occurs when an admin performs system operations.             |
| Saving Account         | Represents the savings account of a member, including balance, loan balance, share capital, and fixed savings.  | Account, Savings             | Occurs when a member opens or manages their savings account. |
| Transaction            | Records transactions related to a member's saving account, such as deposits or payments for loans and programs. | Payment, Deposit             | Occurs during savings account operations.                    |
| Financial Report       | Represents monthly or annual financial reports generated based on saving account operations.                    | Report, Statement            | Occurs when a member requests a financial report.            |
| Application            | Represents an application for membership, loans, or programs by a member.                                       | Request, Form                | Occurs when a member submits an application.                 |
| Membership Application | Represents an application for membership, including status and approval date.                                   | Membership Request           | Occurs when a user applies for membership.                   |
| Loan Application       | Represents a loan application, including loan type, amount, period, and approval details.                       | Loan Request                 | Occurs when a member applies for a loan.                     |
| Program Application    | Represents an application to participate in a program, including description and approval date.                 | Program Request              | Occurs when a member applies for a program.                  |
| System Info            | Represents system-level information, such as loan policies, membership requirements, and interest rates.        | System Settings, System Info | Occurs when an admin updates or views system configurations. |



### 3.3.2 Identify Relationship Types

| Entity Name         | Multiplicity | Relationship     | Entity Name     | Multiplicity |
|---------------------|--------------|------------------|-----------------|--------------|
| Member              | 1...1        | Manage           | Account         | 1...*        |
|                     | 1...1        | Apply            | Application     | 1...*        |
|                     | 1...1        | Request          | FinancialReport | 1...*        |
| Admin               | 1...*        | Approve          | Application     | 1...*        |
| Program Application | 1...1        | Pending          | ProgramStatus   | 1...*        |
| LoanApplication     | 1...1        | Pending          | LoanStatus      | 1...*        |
| Account             | 1...1        | Deposit/Withdraw | SavingAccount   | 1...*        |
|                     | 1...1        | Deposit/Withdraw | SavingAccount   | 1...*        |
|                     | 1...1        | Make             | Transaction     | 1...*        |

### 3.3.3 Attribute Description

#### Entity: Member

| Attributes | Data Type & Length | Description                    | Null | Multi-valued | Example        |
|------------|--------------------|--------------------------------|------|--------------|----------------|
| memberID   | int                | Uniquely identifies the member | No   | No           | 1              |
| username   | Varchar (50)       | Username for member login      | No   | No           | "Audrey"       |
| password   | Varchar (50)       | Password for member login      | No   | No           | "abcabc"       |
| name       | Varchar (100)      | Full name of the member        | No   | No           | "Audrey Sim"   |
| IDNumber   | Varchar (20)       | Identification number          | No   | No           | "A123456789"   |
| gender     | Varchar (10)       | Gender of the member           | No   | No           | "Female"       |
| religion   | Varchar (20)       | Religion of the member         | yes  | No           | "Christianity" |

|                 |                |                                    |     |    |                     |
|-----------------|----------------|------------------------------------|-----|----|---------------------|
| race            | Varchar (20)   | Race of the member                 | Yes | No | “Asian”             |
| maritalStatus   | Varchar (20)   | Marital status of the member       | Yes | No | “Single”            |
| address         | text           | Address of the member              | No  | No | “123 Street Name”   |
| phoneNumber     | Varchar (15)   | Phone number of the member         | No  | No | +60123456789        |
| email           | Varchar (50)   | Email address of the member        | No  | No | “abcabc@gmail.com”  |
| position        | Varchar (50)   | Job position of the member         | No  | No | “Manager”           |
| positionGrade   | Varchar (10)   | Grad level of the position         | Yes | No | “G7”                |
| officeAddress   | text           | Office address of the member       | No  | No | “456 Office Street” |
| MonthlySalary   | Decimal (10,2) | Monthly salary of the member       | No  | No | 3500.50             |
| applicationDate | date           | Date of application                | No  | No | 2025-01-03          |
| approvalStatus  | Varchar (20)   | Approval status of the application | No  | No | “Approved”          |

### Entity: Admin

| Attribute | Data Type & Length | Description                   | Null | Multi-valued | Example             |
|-----------|--------------------|-------------------------------|------|--------------|---------------------|
| adminID   | int                | Uniquely identifies the admin | No   | No           | 1                   |
| username  | Varchar (50)       | Username for admin login      | No   | No           | “admin01”           |
| password  | Varchar (50)       | Password for admin login      | No   | No           | “admin123”          |
| name      | Varchar (100)      | Full name of the admin        | No   | No           | “John Danny”        |
| lastLogin | datetime           | Last login time of the admin  | Yes  | No           | 2024-12-25 14:00:00 |

**Entity: Account**

| Attribute           | Data Type & Length | Description                           | Null | Multi-valued | Example             |
|---------------------|--------------------|---------------------------------------|------|--------------|---------------------|
| accountID           | int                | Uniquely identifies the account       | No   | No           | 101                 |
| memberID            | int                | Foreign key linking to Member entity  | No   | No           | 1                   |
| accountStatus       | Varchar (20)       | Status of the account                 | No   | No           | “Active”            |
| accountCreationDate | datetime           | Date and time the account was created | No   | No           | 2024-05-05 19:00:00 |

**Entity: SavingAccount**

| Attribute       | Data Type & Length | Description                               | Null | Multi-valued | Example             |
|-----------------|--------------------|-------------------------------------------|------|--------------|---------------------|
| balance         | Decimal (10,2)     | Current balance of the savings account    | No   | No           | 5000.00             |
| shareCapital    | Decimal (10,2)     | Share capital amount                      | Yes  | No           | 200.00              |
| fixedSavings    | Decimal (10,2)     | Fixed savings amount                      | Yes  | No           | 1000.00             |
| interestRate    | Decimal (5,2)      | Interest rate for savings                 | Yes  | No           | 2.50                |
| withdrawalLimit | Decimal (5,2)      | Maximum amount of withdrawal allowed      | No   | No           | 5000.00             |
| lastUpdated     | datetime           | Last update timestamp for savings account | No   | No           | 2024-12-25 12:00:00 |

**Entity: LoanAccount**

| Attribute       | Data Type & Length | Description                            | Null | Multi-valued | Example             |
|-----------------|--------------------|----------------------------------------|------|--------------|---------------------|
| loanBalance     | Decimal (10,2)     | Outstanding loan balance               | No   | No           | 10000.00            |
| overdueAmount   | Decimal (10,2)     | Amount overdue on the loan             | Yes  | No           | 500.00              |
| penaltyRate     | Decimal (5,2)      | Penalty rate for overdue payments      | Yes  | No           | 1.50                |
| nextPaymentRate | date               | Date of next loan payment              | No   | No           | 2025-02-02          |
| lastUpdated     | datetime           | Last update timestamp for loan account | No   | No           | 2024-12-12 12:00:00 |

**Entity: Transaction**

| Attribute         | Data Type & Length | Description                                   | Null | Multi-valued | Example             |
|-------------------|--------------------|-----------------------------------------------|------|--------------|---------------------|
| transactionID     | int                | Uniquely identifies the transaction           | No   | No           | 1                   |
| accountID         | int                | Foreign key linking to Account entity         | No   | No           | 101                 |
| transactionType   | Varchar (20)       | Type of transaction (e.g., Deposit, Withdraw) | No   | No           | “Deposit”           |
| amount            | Decimal (10,2)     | Transaction amount                            | No   | No           | 1000.00             |
| transactionDate   | datetime           | Date and time of the transaction              | No   | No           | 2024-12-12 10:00:00 |
| transactionStatus | Varchar (20)       | Status of the transaction (e.g., Completed)   | No   | No           | “Completed”         |

**Entity: FinancialReport**

| Attribute     | Data Type & Length | Description                                        | Null | Multi-valued | Example             |
|---------------|--------------------|----------------------------------------------------|------|--------------|---------------------|
| reportID      | int                | Uniquely identifies the financial report           | No   | No           | 1                   |
| memberID      | int                | Foreign key linking to Member entity               | No   | No           | 1                   |
| reportType    | Varchar (50)       | Type of the financial report (e.g., Savings, Loan) | No   | No           | “Savings Report”    |
| generatedDate | datetime           | Date and time the report was generated             | No   | No           | 2024-12-25 15:00:00 |
| details       | text               | Detailed description of the report                 | Yes  | No           | “Report content”    |

**Entity: Application**

| Attribute       | Data Type & Length | Description                          | Null | Multi-valued | Example    |
|-----------------|--------------------|--------------------------------------|------|--------------|------------|
| applicationID   | int                | Uniquely identifies the application  | No   | No           | 1          |
| memberID        | int                | Foreign key linking to Member entity | No   | No           | 1          |
| applicationDate | date               | Date the application was submitted   | No   | No           | 2024-12-12 |

**Entity: LoanApplication**

| Attribute          | Data Type & Length | Description                               | Null | Multi-valued | Example     |
|--------------------|--------------------|-------------------------------------------|------|--------------|-------------|
| loanType           | Varchar (50)       | Type of loan applied                      | No   | No           | “Personal”  |
| loanAmount         | Decimal (10,2)     | Amount of loan applied                    | No   | No           | 40000.00    |
| loanPeriod         | int                | Duration of the loan in months            | No   | No           | 60          |
| monthlyInstallment | Decimal (10,2)     | Monthly repayment amount                  | No   | No           | 1000.00     |
| bankAccNo          | Varchar (30)       | Bank account number for loan disbursement | No   | No           | “123456789” |
| bankName           | Varchar (50)       | Name of the bank                          | No   | No           | “Maybank”   |
| approvalDate       | date               | Date the loan was approved                | Yes  | No           | 2024-12-31  |

**Entity: ProgramApplication**

| Attribute          | Data Type & Length | Description                               | Null | Multi-valued | Example              |
|--------------------|--------------------|-------------------------------------------|------|--------------|----------------------|
| programName        | Varchar (80)       | Name of the program applied               | No   | No           | “Financial Training” |
| programDescription | text               | Description of the program                | Yes  | No           | “Program details”    |
| programFee         | Decimal (10,2)     | Fee of the program                        | No   | No           | 200.00               |
| approvalDate       | date               | Date the program application was approved | Yes  | No           | 2024-11-15           |

**Entity: ProgramStatus**

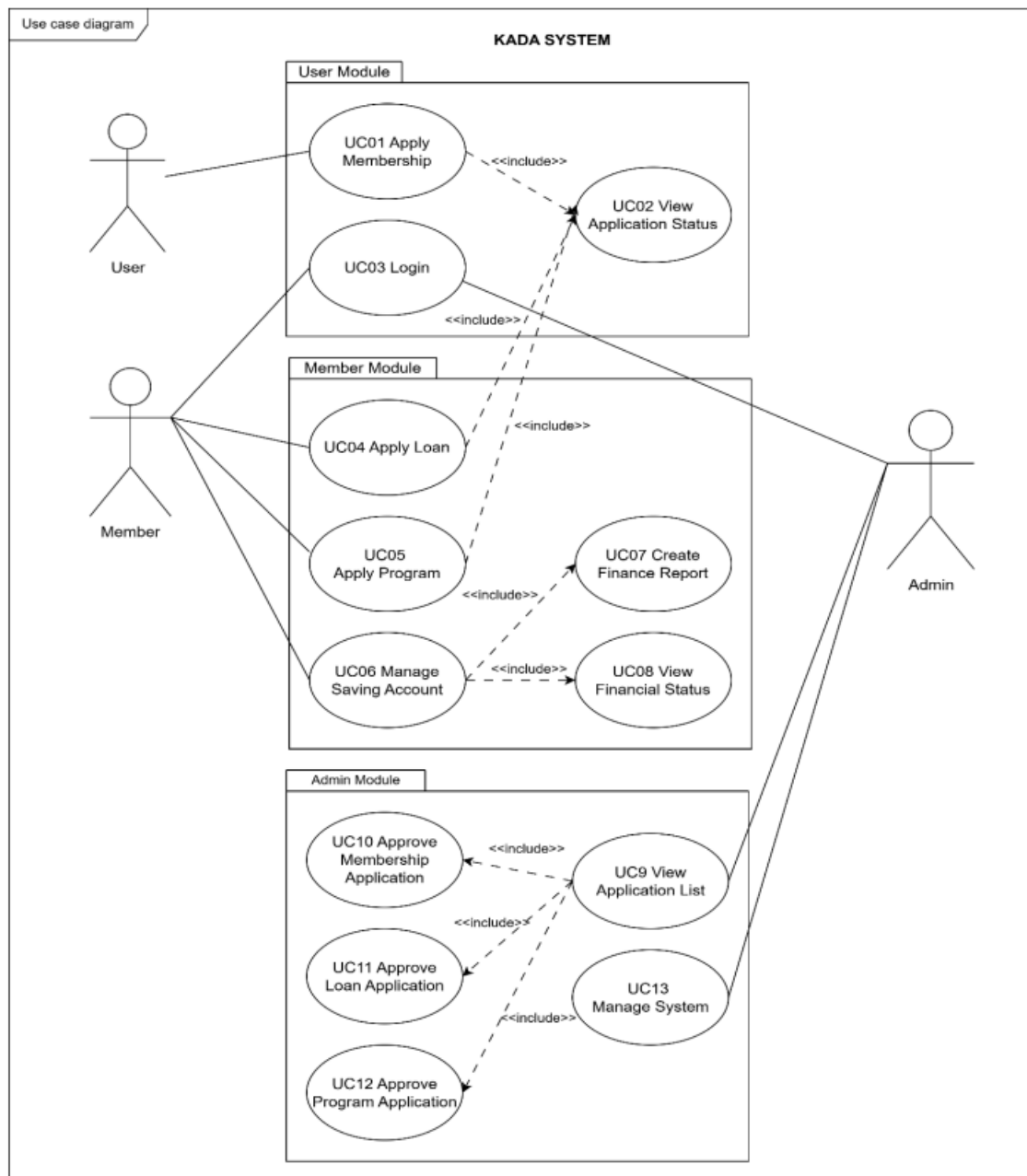
| Attribute       | Data Type & Length | Description                                 | Null | Multi-valued | Example             |
|-----------------|--------------------|---------------------------------------------|------|--------------|---------------------|
| statusID        | int                | Uniquely identifies the program status      | No   | No           | 1                   |
| applicationID   | int                | Foreign key linking to Application entity   | No   | No           | 1                   |
| savingAccountID | int                | Foreign key linking to SavingAccount entity | No   | No           | 101                 |
| statusType      | Varchar (50)       | Type of the status                          | No   | No           | “Pending”           |
| statusRemarks   | text               | Remarks or comments on the status           | Yes  | No           | “Approved by admin” |
| statusTimestamp | datetime           | Date and time of the status update          | No   | No           | 2025-01-01 16:00:00 |

**Entity: LoanStatus**

| Attribute       | Data Type & Length | Description                               | Null | Multi-valued | Example             |
|-----------------|--------------------|-------------------------------------------|------|--------------|---------------------|
| statusID        | int                | Uniquely identifies the loan status       | No   | No           | 1                   |
| applicationID   | int                | Foreign key linking to Application entity | No   | No           | 1                   |
| loanAccountID   | int                | Foreign key linking to LoanAccount entity | No   | No           | 201                 |
| statusType      | Varchar (50)       | Type of the status                        | No   | No           | “Pending”           |
| statusRemarks   | text               | Remarks or comments on the status         | Yes  | No           | “Awaiting review”   |
| statusTimestamp | datetime           | Date and time of the status update        | No   | No           | 2025-01-01 16:00:00 |

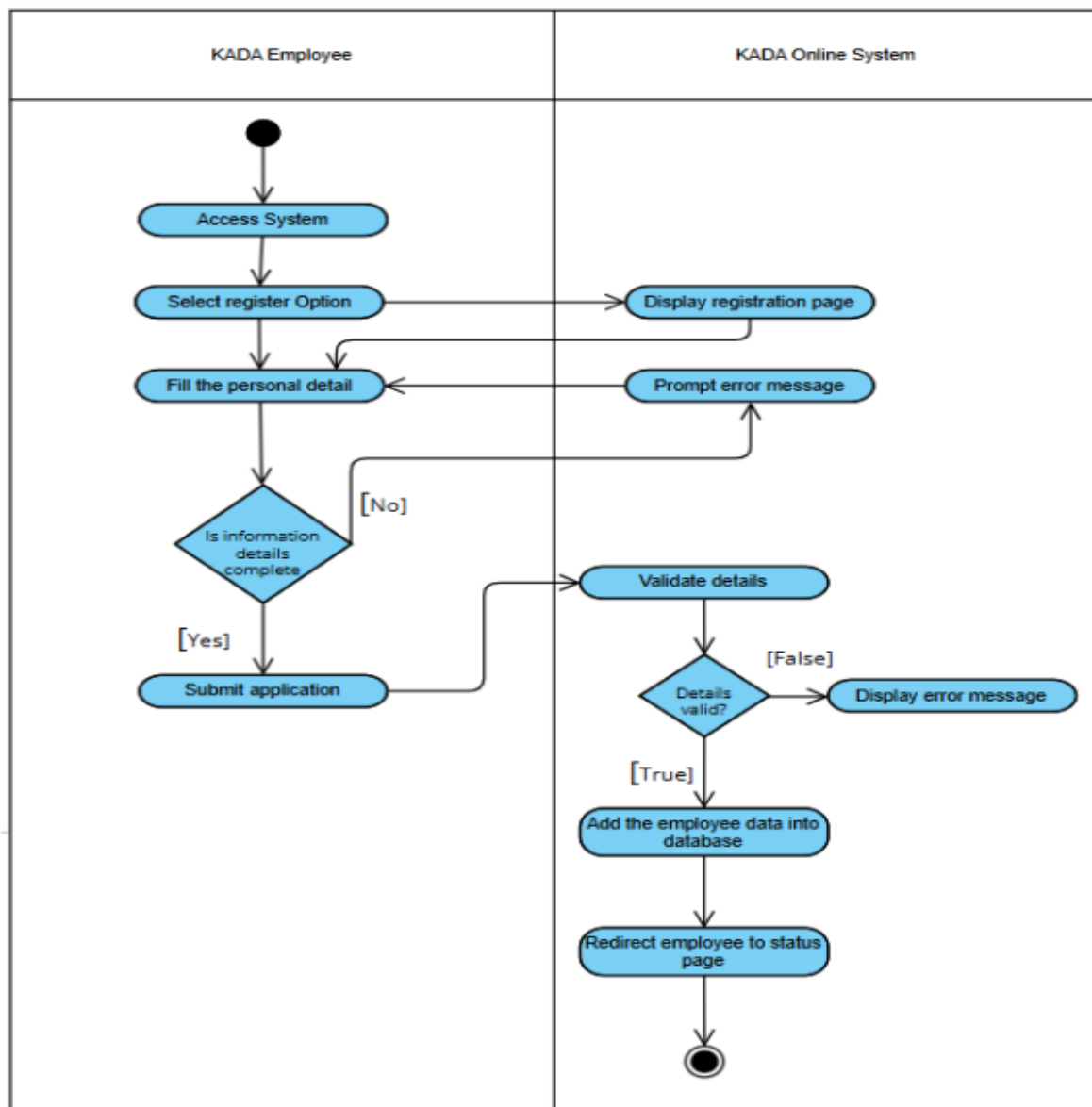
## Appendices

### 4.1 Use Case Diagram for the system.

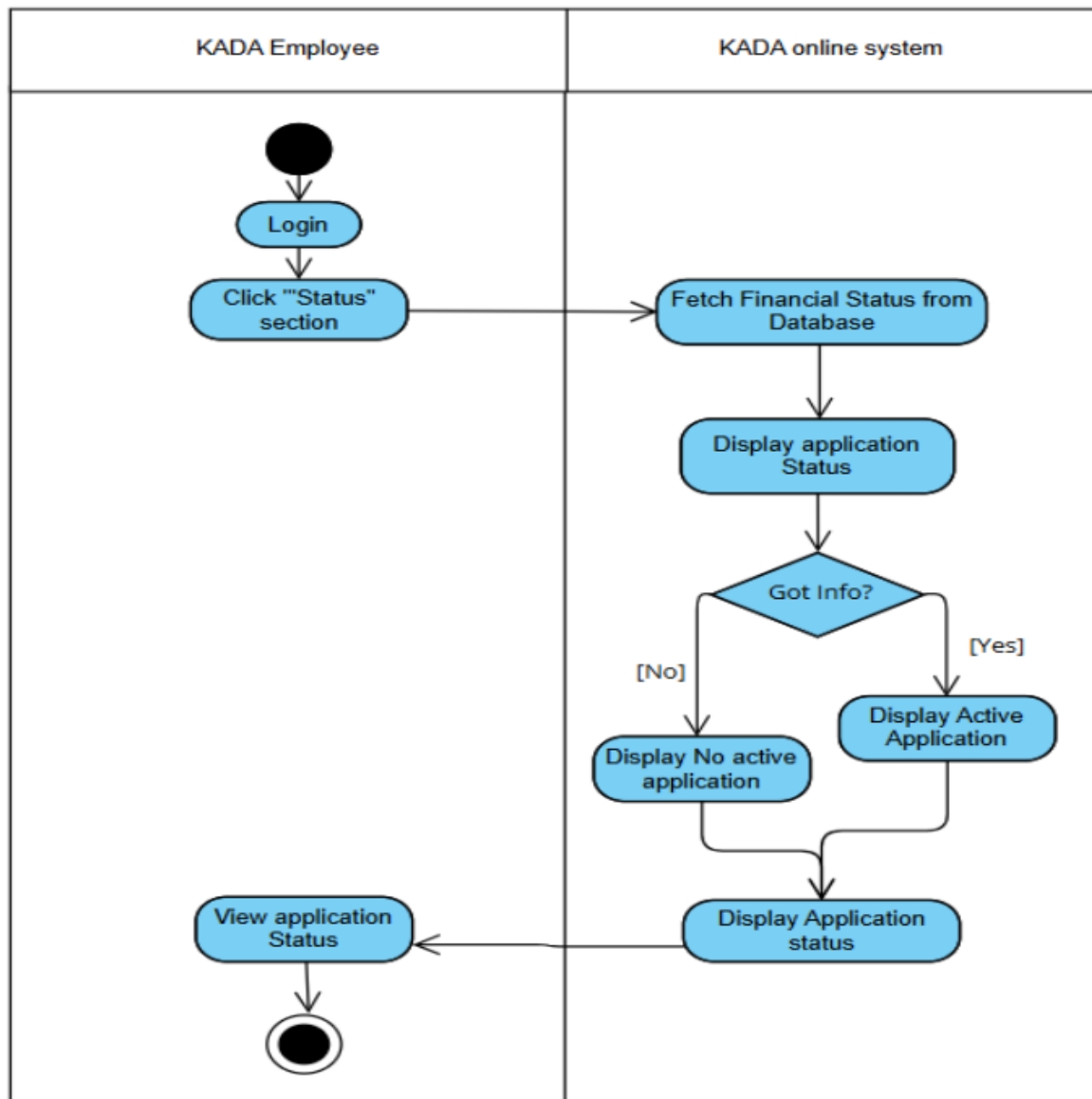




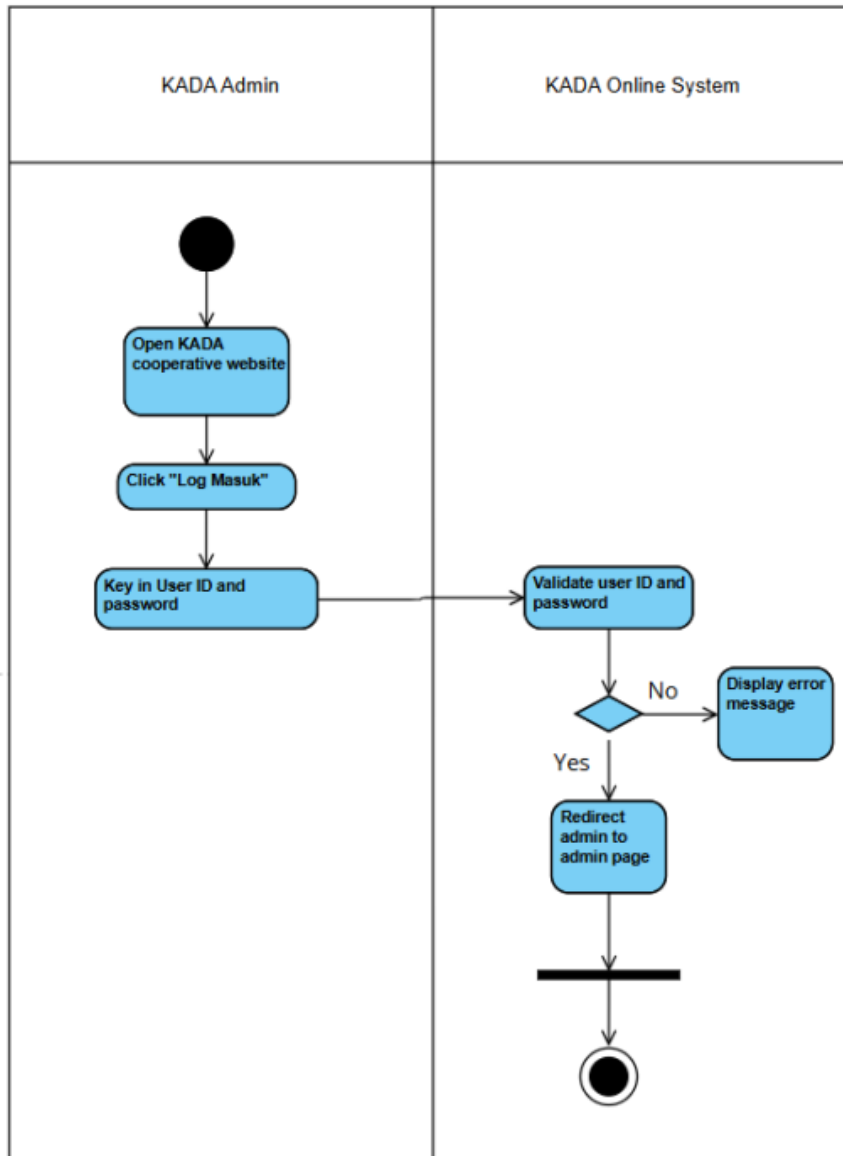
## 4.2 Activity Diagrams for every use case identified.



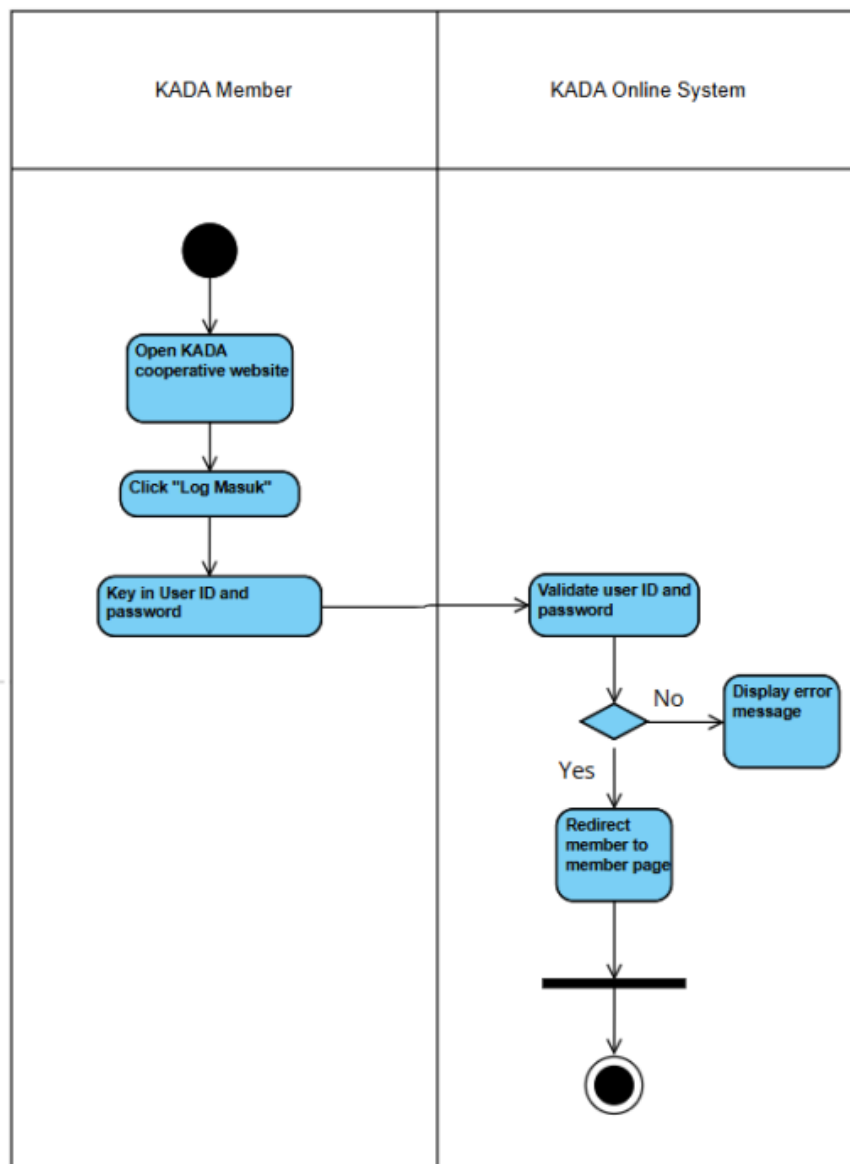
**Figure 4.2.1 UC01: Apply Membership**



**Figure 4.2.2 UC02: View Application Status**



**Figure 4.2.3 UC03: Log in (Admin)**



**Figure 4.2.3 UC03: Log In (Member)**

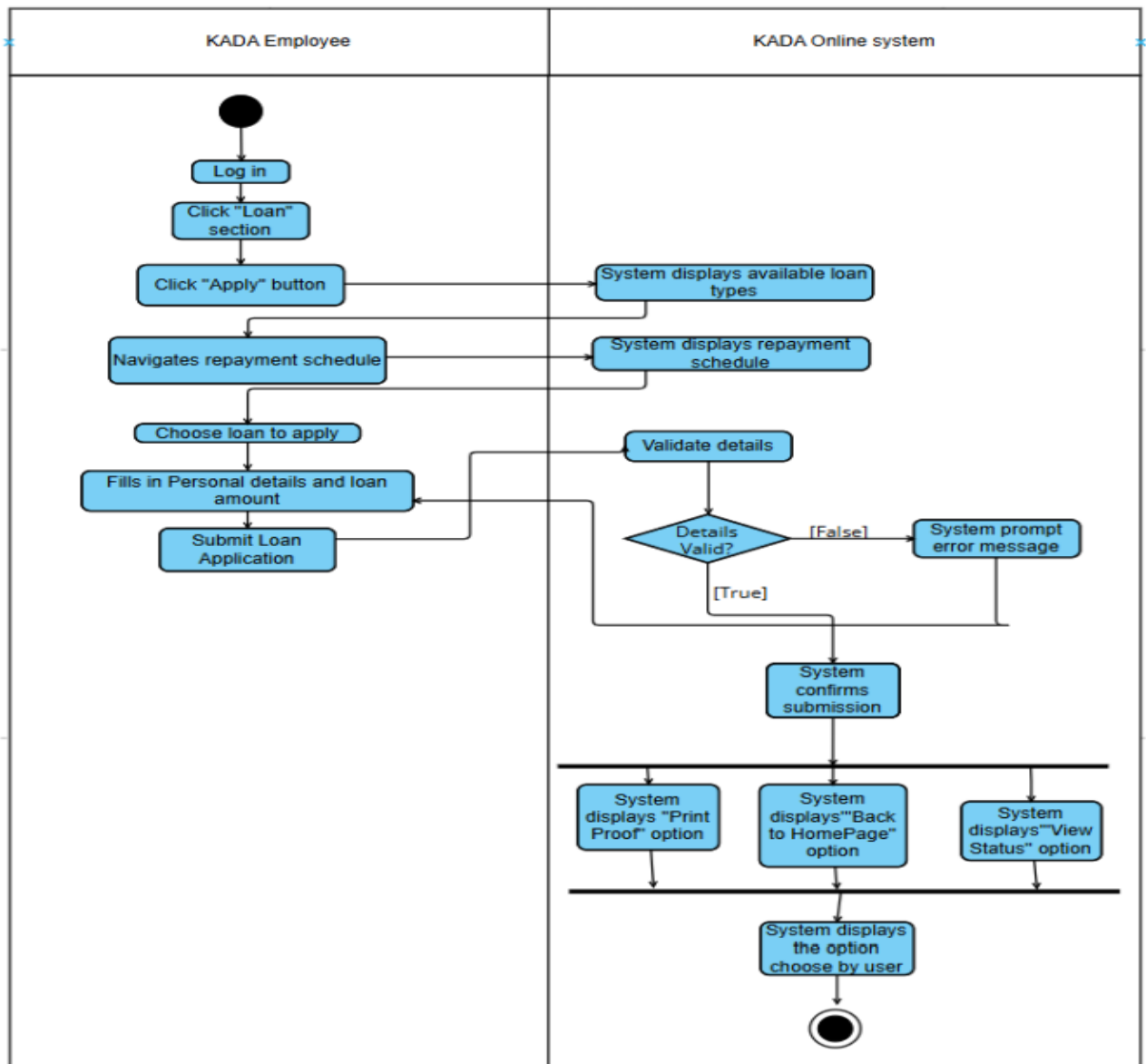


Figure 4.2.4 UC04: Apply Loan

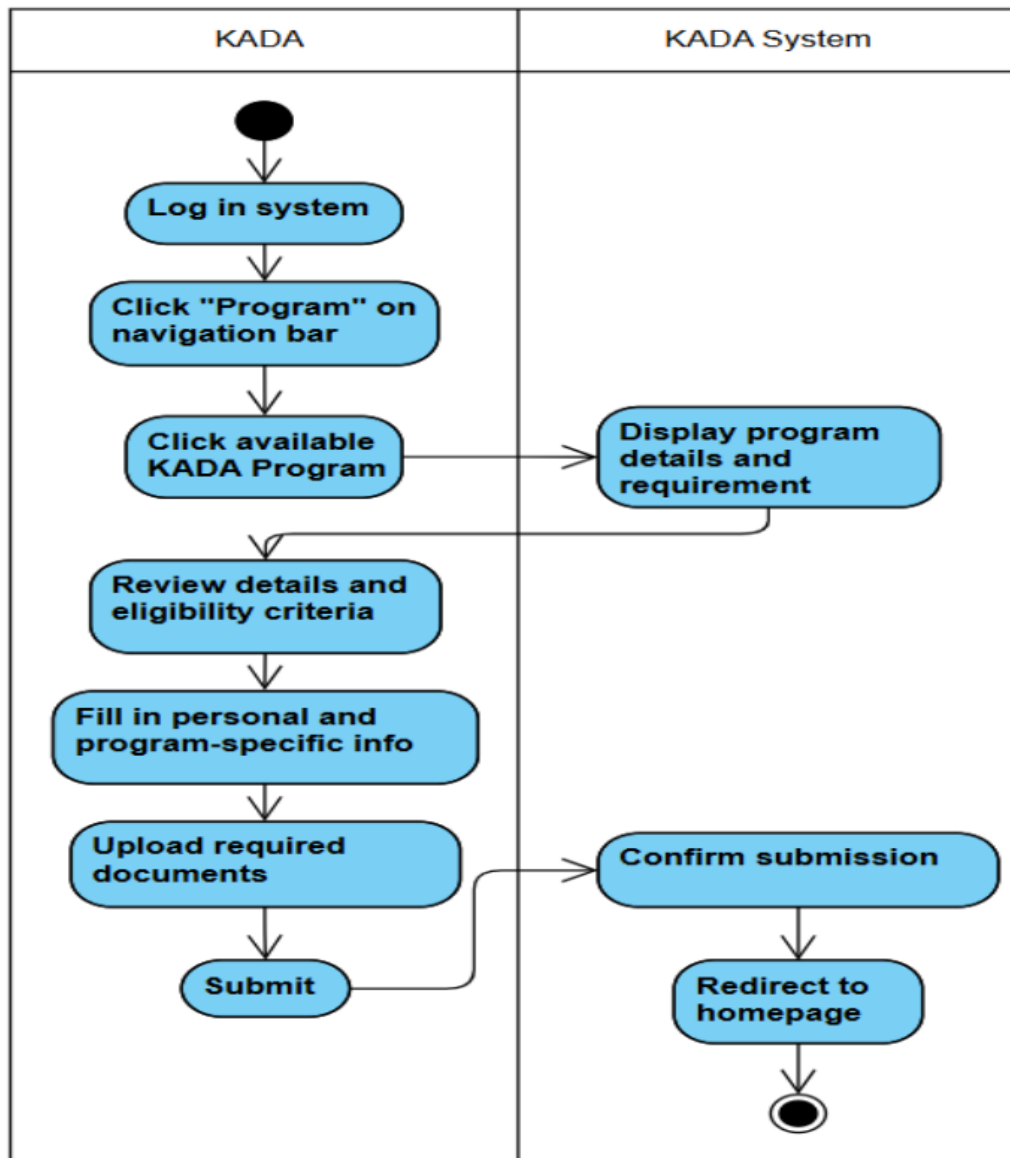
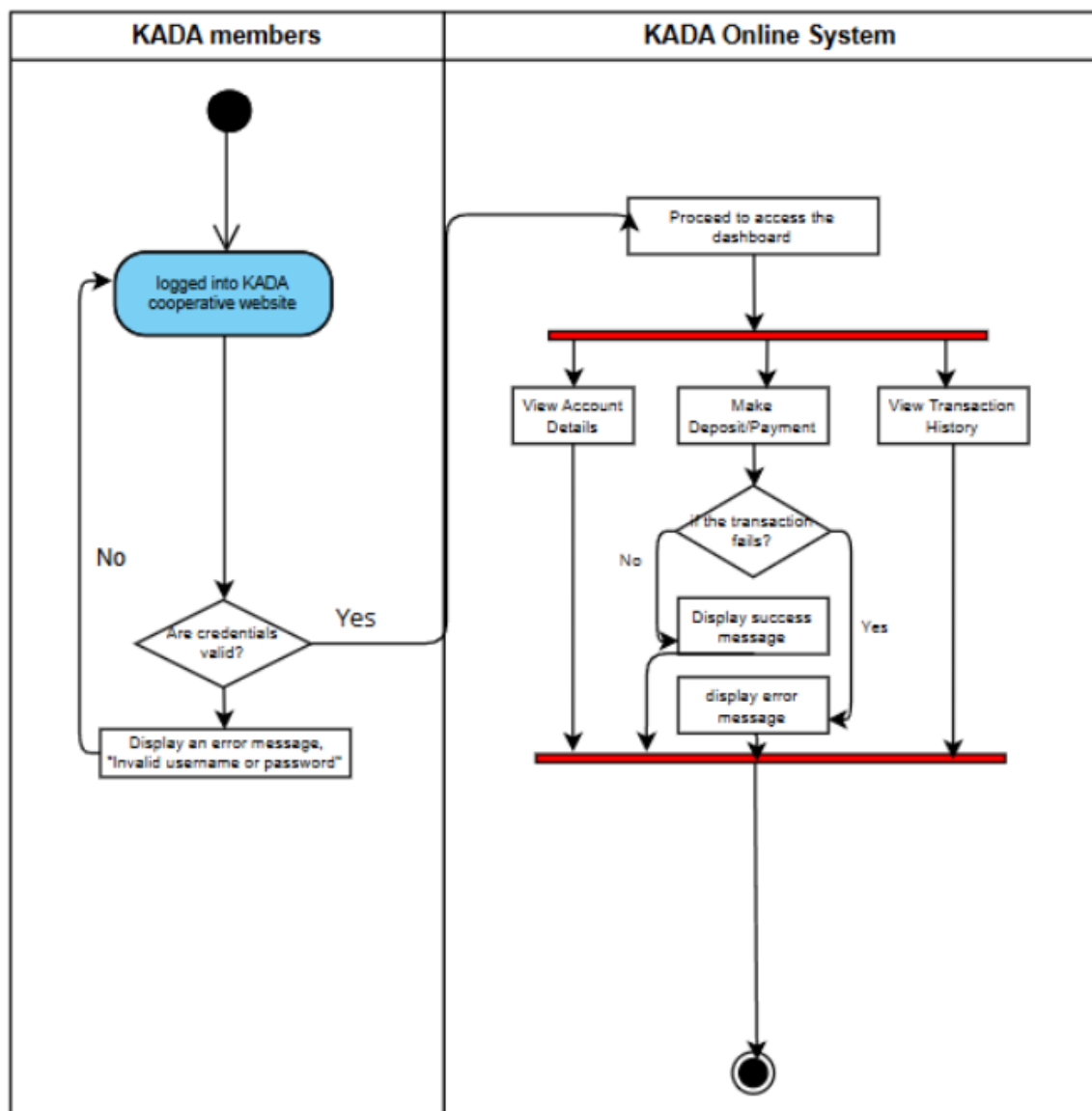
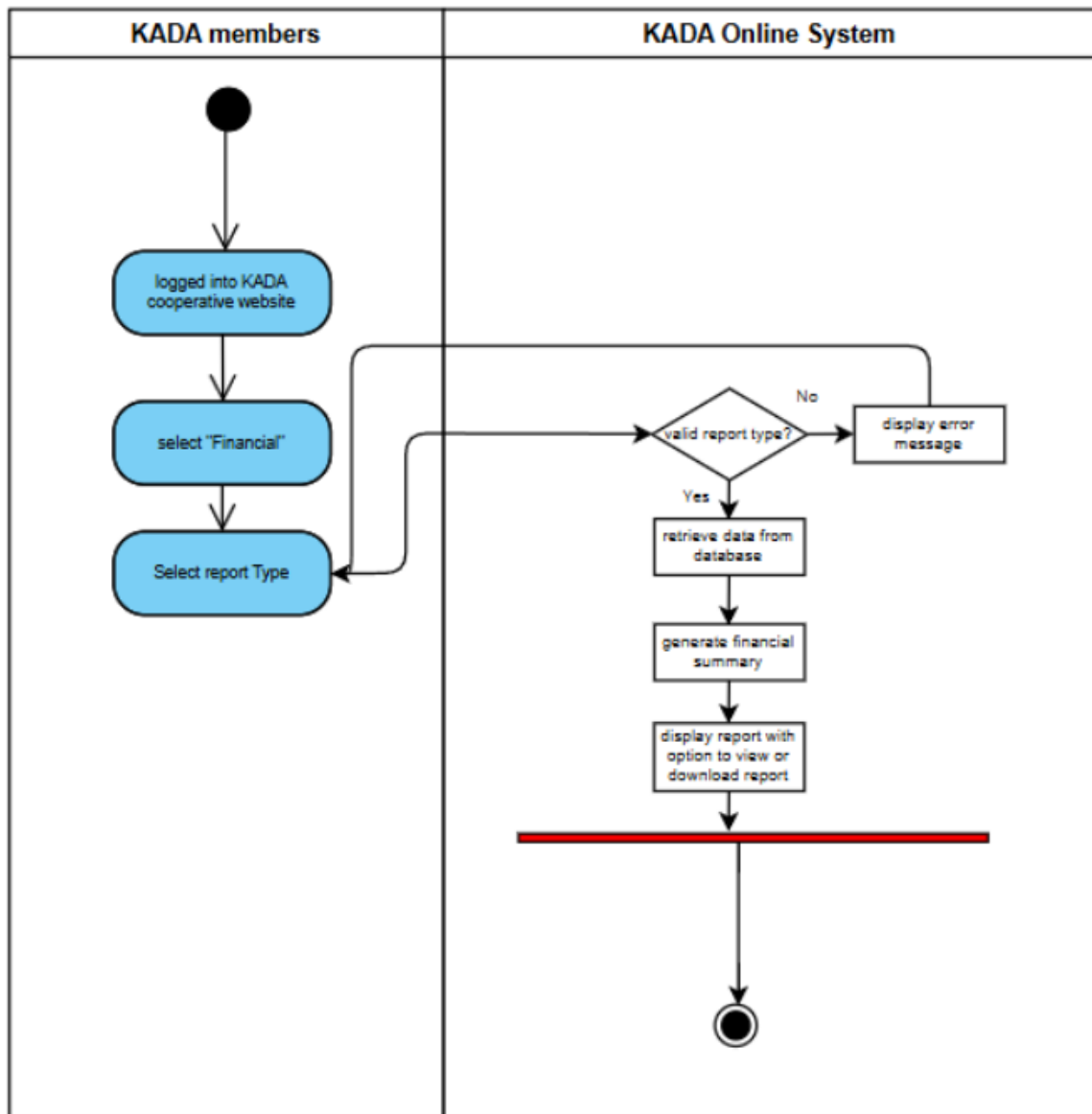


Figure 4.2.5 UC05: Apply Program

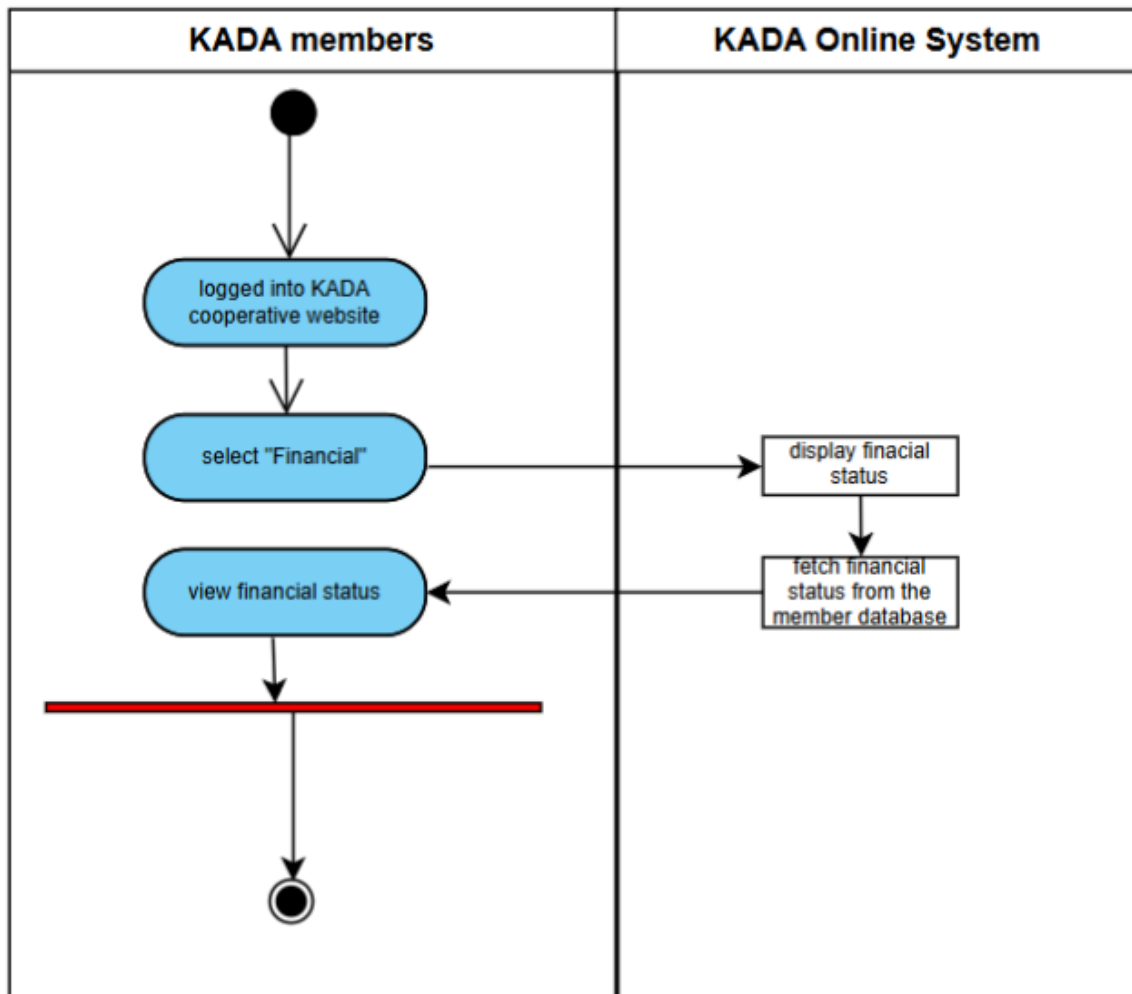


**Figure 4.2.6 UC06: Manage Savings Account**

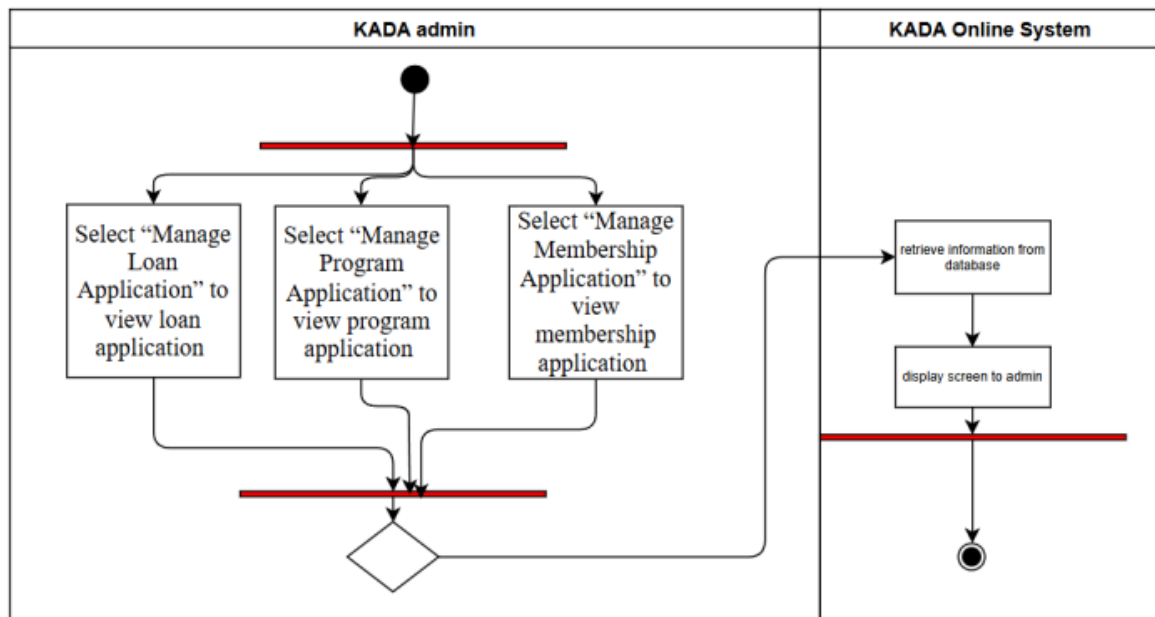


**Figure 4.2.7 UC07: Create Financial Report**

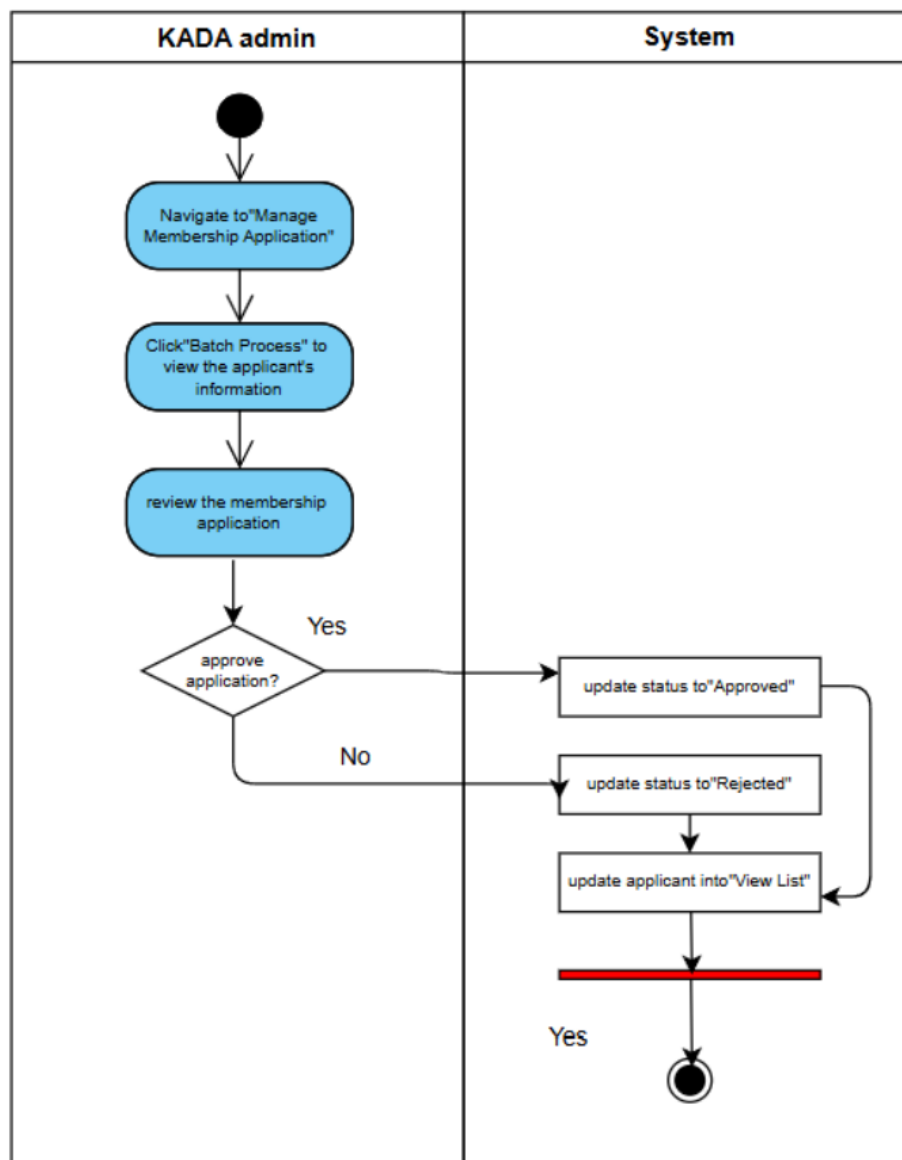




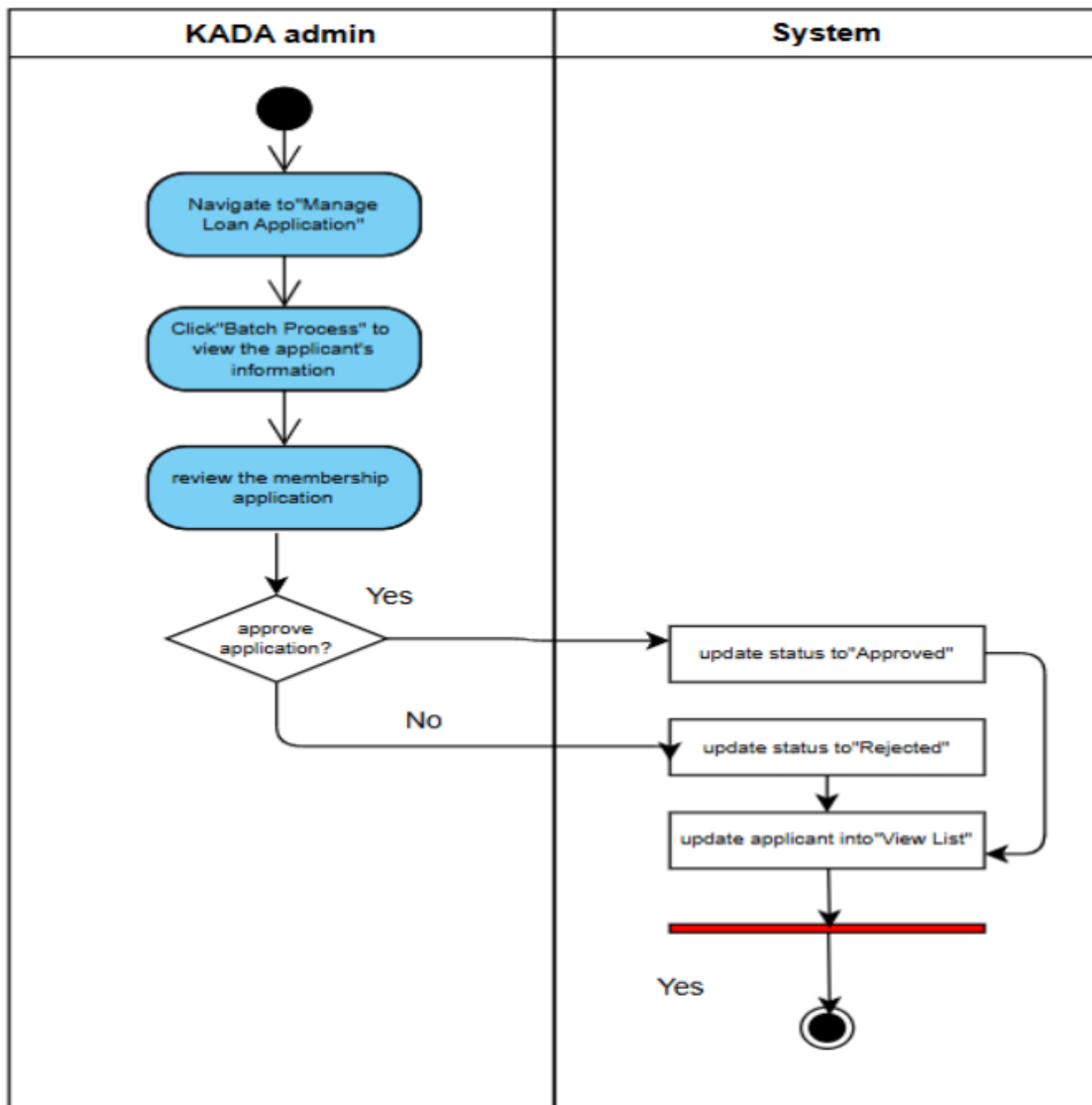
**Figure 4.2.8 UC08: View Financial Status**



**Figure 4.2.9 UC09: Admin View Application List**



**Figure 4.2.10 UC10: Approve Membership Application**



**Figure 4.2.11 UC11: Approve Loan Application**

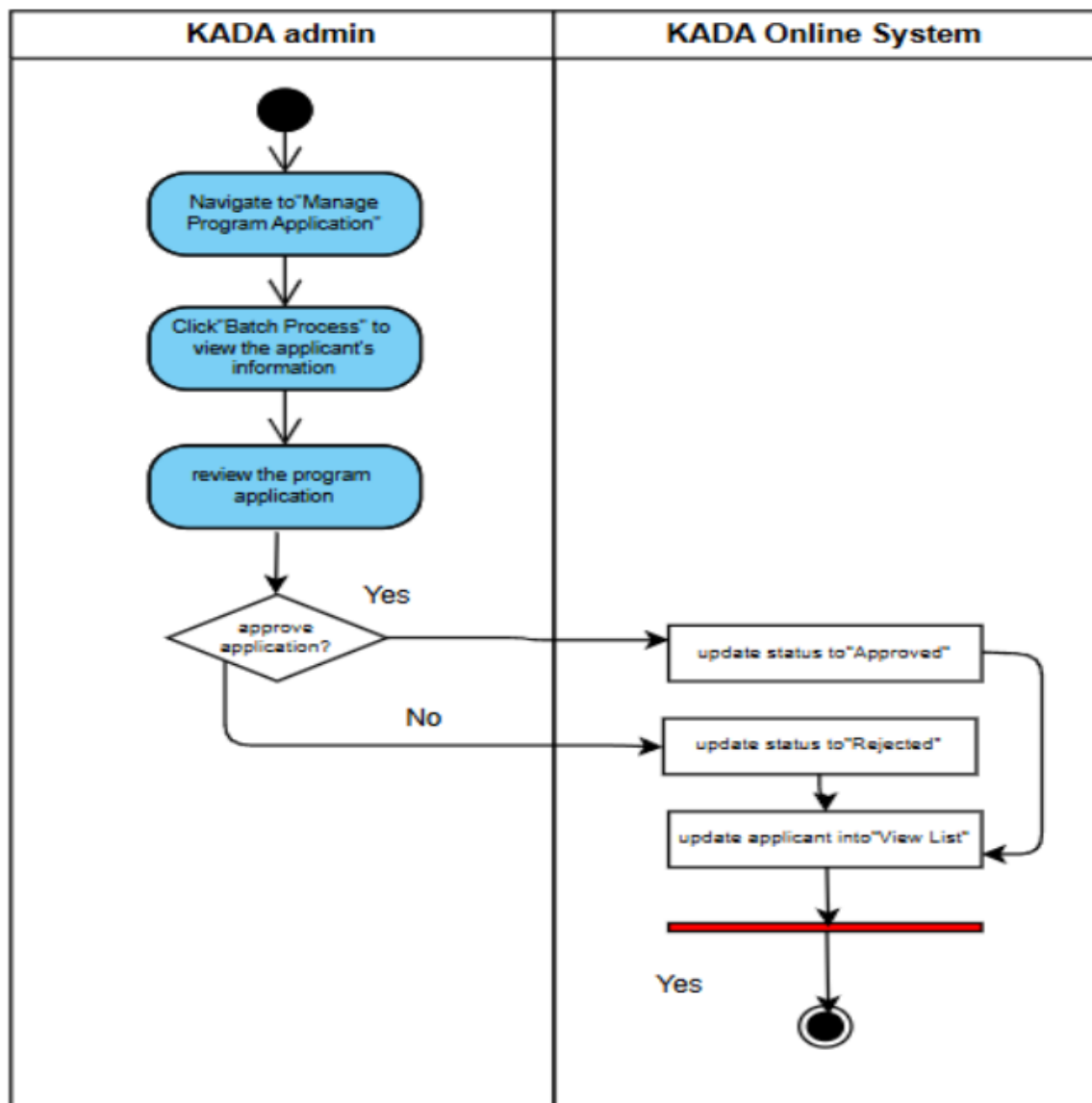


Figure 4.2.12 UC12: Approve Program Application

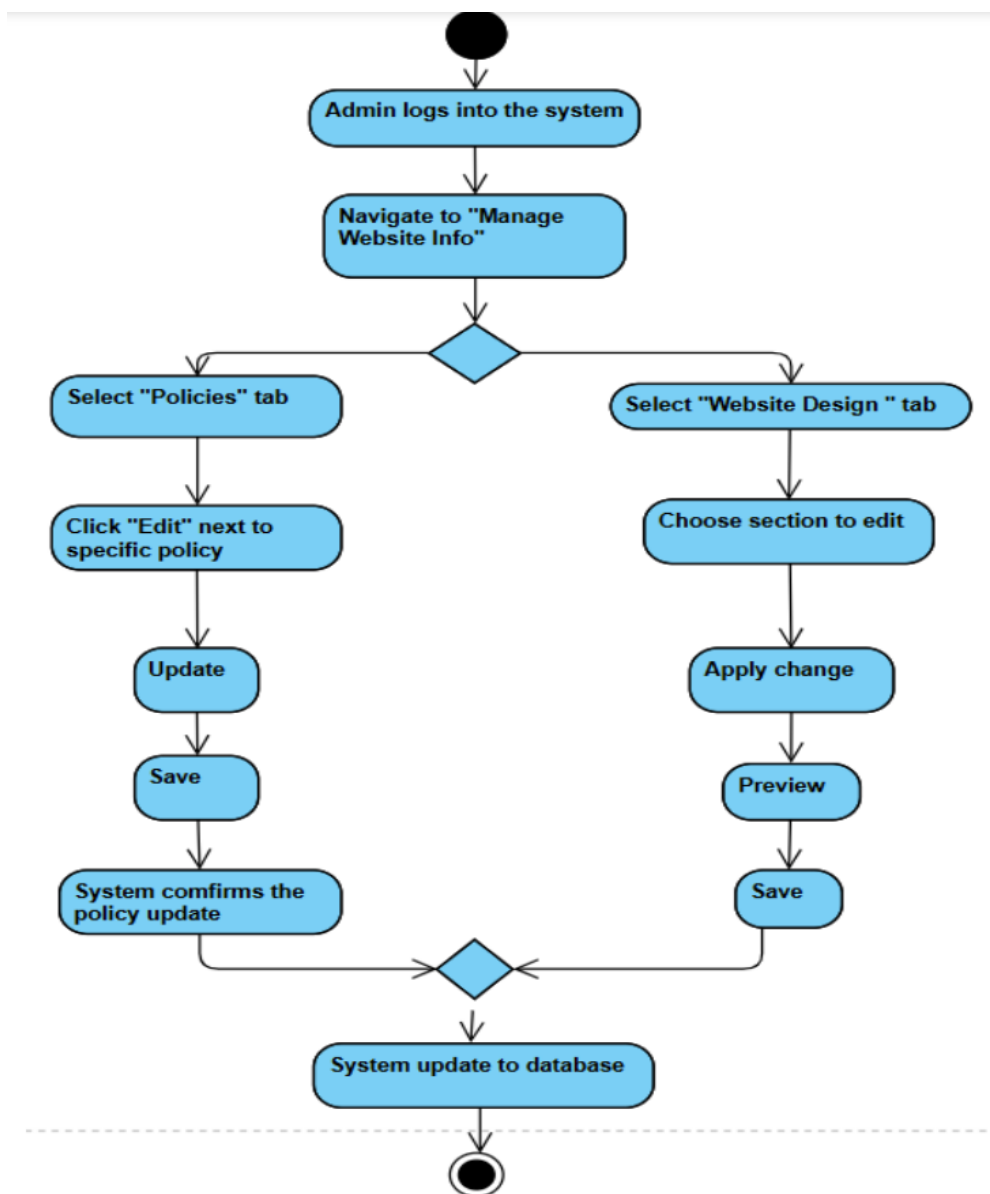
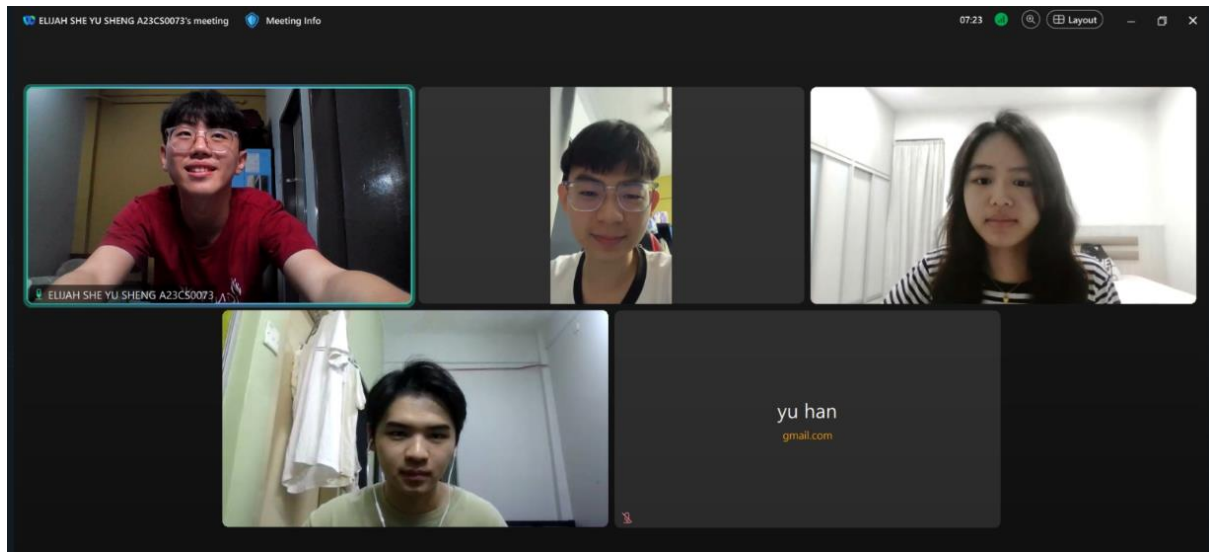
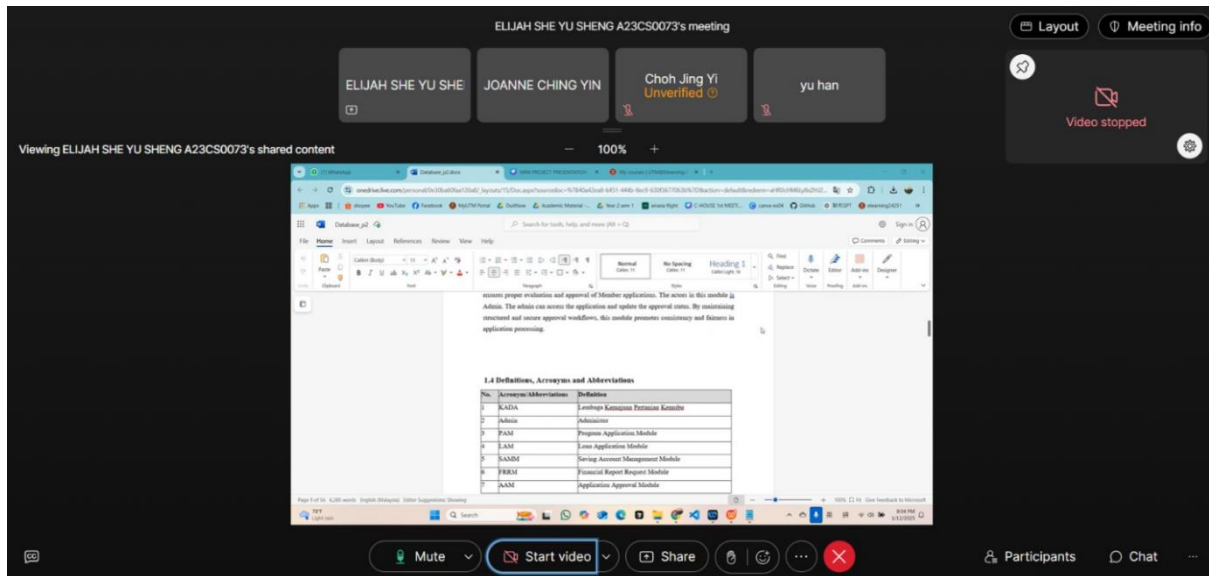


Figure 4.2.13 UC13: Manage Website

### 4.3 Meeting Log



We initiated our meeting at 9:00 a.m. on 28 December 2024, with the aim of conducting a group discussion focused on the second phase of the database project. All group members were present and contributed actively and we focus on task distribution and brainstorming ideas to achieve the project's objectives efficiently.



We concluded our meeting at 11:30 a.m. on 9 January 2025, with active participation from every group member who dedicated their utmost effort to finalize the phase 2 project. Each team member shared updates on their respective contributions to ensure the seamless progress of our project.